

Reductive groups: uniqueness of pinned groups

SGA 3, Exposé XXIII

English draft

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About this draft

This is an unofficial English translation of Exposé XXIII, “Reductive groups: uniqueness of pinned groups” from *Schémas en groupes* (SGA 3). The source is the recomposition edited by P. Gille and P. Polo, PDF `Exp23-13oct24.pdf` (37 pages), from <https://webusers.imj-prg.fr/~patrick.polo/SGA3/>. The seminar was directed by M. Demazure and A. Grothendieck (IHÉS, 1962–1964).

Statement numbers follow that PDF. A clear typographical slip is corrected in the English and marked with a `% typo:` comment; such slips are listed in the README. A slip that would change the mathematics is left as printed and marked `% typo?:`. Scholarly proofreading remains outstanding.

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Exposé XXIII. Reductive groups: uniqueness of pinned groups

⁽⁰⁾The aim of this exposé is to prove the uniqueness theorem (Theorem 4.1). This was proved by Chevalley in the case of an algebraically closed field; the method of reduction to rank two used here is also due to Chevalley (see *Bible*, exp. 23 and 24). Along the way, we obtain an explicit description of reductive groups by generators and relations (3.5).

1. Pinnings

Definition 1.1. Let S be a scheme, and (G, T, M, R) a split S -group (XXII, 1.13). A *pinning*⁽¹⁾ of this split group is the datum of a system Δ of *simple roots* of R and, for each $\alpha \in \Delta$, a section $X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times$.

In other words, a pinning of the reductive group G over the nonempty scheme S is the datum:

- (i) of a maximal torus T ,
- (ii) of an abelian group M and an isomorphism $T \simeq D_S(M)$,
- (iii) of a root system R of G with respect to T ,
- (iv) of a system of simple roots Δ of R ,
- (v) of an $X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times$ for every $\alpha \in \Delta$, that is, of a

$$u_\alpha = \exp_\alpha(X_\alpha) \in U_\alpha^\times(S) \quad \text{for every } \alpha \in \Delta,$$

satisfying condition (D 1) of Exp. XXII, 1.13 (indeed, condition (D 2) of *loc. cit.* is automatically satisfied⁽²⁾).

Every split group has a pinning; in particular, every reductive group is locally pinnable for the étale topology.

1.2. — If G is a *pinned* S -group, that is, a split S -group equipped with a pinning, it is canonically equipped with the system of positive roots R_+ defined by Δ , the corresponding Borel subgroup $B = B_{R_+}$, the opposite Borel subgroup $B^- = B_{R_-}$, the unipotent groups $U = B^u$, $U^- = (B^-)^u$, the open subset $U^- \cdot T \cdot U$, etc. Likewise, for each $\alpha \in \Delta$, one has a canonical isomorphism of vector groups

$$p_\alpha : \mathbf{G}_{a,S} \xrightarrow{\sim} U_\alpha, \quad x \longmapsto \exp_\alpha(xX_\alpha) = u_\alpha^x,$$

normalized by T with multiplier α , and whose datum is equivalent to that of X_α (Exp. XXII, 1.1).

By duality, one deduces from it an $X_{-\alpha} \in \Gamma(S, \mathfrak{g}^{-\alpha})^\times$ and an isomorphism

$$p_{-\alpha} : \mathbf{G}_{a,S} \xrightarrow{\sim} U_{-\alpha}$$

⁽⁰⁾**N.D.E.** Version of 13/10/2024.

⁽¹⁾**N.D.E.** Demazure tells us that behind this terminology lies the image of a butterfly (which Grothendieck supplied to him): the body is a maximal torus T , the wings are two opposite Borel subgroups with respect to T ; one splits the butterfly by spreading its wings, then fixes elements in the additive groups (pins) to rigidify the situation (i.e., to eliminate automorphisms).

⁽²⁾**N.D.E.** It is implied by condition (v), i.e., the existence of a section $X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times$.

which is the contragredient of the preceding one (Exp. XXII, 1.3). We shall put (Exp. XX, 3.1)

$$\begin{aligned} w_\alpha &= w_\alpha(X_\alpha) = p_\alpha(1)p_{-\alpha}(-1)p_\alpha(1) \\ &= p_{-\alpha}(-1)p_\alpha(1)p_{-\alpha}(-1). \end{aligned}$$

One then has (*loc. cit.* 3.1, 3.7)

$$\begin{aligned} w_\alpha^2 &= \alpha^*(-1), \quad \text{int}(w_\alpha)t = s_\alpha(t) = t \cdot \alpha^*(\alpha(t)^{-1}), \\ \begin{cases} \text{int}(w_\alpha)p_\alpha(x) = p_{-\alpha}(-x) = p_{-\alpha}(x)^{-1}, \\ \text{Ad}(w_\alpha)X_\alpha = -X_{-\alpha}, \end{cases} \\ \begin{cases} \text{int}(w_\alpha)p_{-\alpha}(x) = p_\alpha(-x) = p_\alpha(x)^{-1}, \\ \text{Ad}(w_\alpha)X_{-\alpha} = -X_\alpha. \end{cases} \end{aligned}$$

We shall systematically use the preceding notation in what follows.

Definition 1.3. Let S be a scheme, and $(G, T, M, R, \Delta, (X_\alpha))$ and $(G', T', M', \dots)$ two pinned S -groups. One says that the morphism of split groups (Exp. XXII, 4.2.1)

$$f : G \longrightarrow G'$$

is compatible with the pinnings, or defines a *morphism of pinned groups*, if the bijection $d : R \longrightarrow R'$ associated with it (cf. *loc. cit.*) satisfies $d(\Delta) = \Delta'$ and if, for every $\alpha \in \Delta$, one has

$$f(\exp_\alpha(X_\alpha)) = \exp_{d(\alpha)}(X'_{d(\alpha)}), \quad \text{i.e.} \quad f(u_\alpha) = u'_{d(\alpha)}.$$

(3)

1.4. — If one denotes by $q(\alpha)$ the integer of *loc. cit.*, one therefore has

$$f(p_\alpha(x)) = p'_{d(\alpha)}(x^{q(\alpha)}) \quad \text{for } \alpha \in \Delta,$$

and hence also

$$f(p_{-\alpha}(x)) = p'_{-d(\alpha)}(x^{q(\alpha)}), \quad f(w_\alpha) = w'_{d(\alpha)}.$$

Recall (Exp. XXII, 4.2) that for every $\alpha \in R$, and all $z \in \mathbf{G}_m(S')$, $t \in T(S')$, one has:

$$\begin{aligned} f(\alpha^*(z)) &= (d(\alpha)^*(z))^{q(\alpha)} = d(\alpha)^*(z^{q(\alpha)}), \\ d(\alpha)(f(t)) &= \alpha(t)^{q(\alpha)}. \end{aligned}$$

1.5. — Let us call a *pinned root datum* a root datum equipped with a system of simple roots, and a *p-morphism of pinned root data* a p -morphism of root data (Exp. XXI, 6.8) taking simple roots to simple roots.

If G is a pinned S -group, one denotes by $\mathcal{R}(G)$ the corresponding pinned root datum (it is the root datum of Exp. XXII, 1.14 equipped with Δ). Let p be the integer defined in Exp. XXII, 4.2.2. One then has

Scholium 1.6. *The correspondence $G \longmapsto \mathcal{R}(G)$ defines a functor from the category of pinned reductive S -groups to that of pinned root data (with p -morphisms as morphisms).*

1.7. — **The pinned groups Z_{Δ_1}**

Let Δ_1 be a subset of the system of simple roots Δ of the pinned group G . Let T_{Δ_1} be the maximal torus of $\bigcap_{\alpha \in \Delta_1} \text{Ker}(\alpha)$; put

$$Z_{\Delta_1} = \text{Centr}_G(T_{\Delta_1}).$$

Write $R' = \mathbf{Z} \cdot \Delta_1 \cap R$; one knows (Exp. XXII, 5.10.7) that Z_{Δ_1} is a reductive S -group with radical T_{Δ_1} , that (T, M, R') is a splitting of it and Δ_1 a system of simple roots. It follows

(3) **N.D.E.** We have denoted the bijection $R \xrightarrow{\sim} R'$ by d (instead of u), to avoid the notation $u'_{u(\alpha)}$.

that $(Z_{\Delta_1}, T, M, R', \Delta_1, (X_\alpha)_{\alpha \in \Delta_1})$ is a pinned S-group. We shall always equip Z_{Δ_1} with this pinning. In particular, one will consider the groups

$$Z_\alpha = Z_{\{\alpha\}}, \quad Z_{\alpha\beta} = Z_{\{\alpha, \beta\}}.$$

One writes $B_{\Delta_1} = B \cap Z_{\Delta_1}$; one knows (*loc. cit.*) that this is the canonical Borel subgroup⁽⁴⁾ of Z_{Δ_1} , and that its unipotent part is $U_{\Delta_1} = U \cap Z_{\Delta_1}$. In particular, one has

$$B_\alpha = T \cdot U_\alpha.$$

We shall write

$$U_{\alpha\beta} = U_{\{\alpha, \beta\}} = U \cap Z_{\alpha\beta} = \prod_{\gamma \in R_{\alpha\beta}^+} U_\gamma,$$

where $R_{\alpha\beta}^+$ denotes the set of positive roots that are linear combinations of α and β .

Now let $f : G \rightarrow G'$ be a morphism of pinned S-groups. If $d : R \rightarrow R'$ is the corresponding bijection and Δ_1 is a subset of Δ , then $d(\Delta_1) = \Delta'_1$ is a subset of Δ' , and it is clear that f sends T_{Δ_1} into $T'_{\Delta'_1}$, hence Z_{Δ_1} into $Z'_{\Delta'_1}$. The corresponding S-morphism:

$$f_{\Delta_1} : Z_{\Delta_1} \rightarrow Z'_{\Delta'_1}$$

is compatible with the canonical pinning; it defines a morphism of pinned root data

$$\mathcal{R}(f_{\Delta_1}) : \mathcal{R}(Z_{\Delta_1}, T, M, \dots) \rightarrow \mathcal{R}(Z'_{\Delta'_1}, T', M', \dots)$$

and the morphisms $M' \rightarrow M$ underlying $\mathcal{R}(f)$ and $\mathcal{R}(f_{\Delta_1})$ coincide.

1.8. — Study of the group $\text{Norm}_G(T)$

For each pair (α, β) of simple roots, denote by $n_{\alpha\beta}$ the order of the element $s_\alpha s_\beta$ of the Weyl group W . In particular, one has $n_{\alpha\alpha} = 1$. Thus one has $(w_\alpha w_\beta)^{n_{\alpha\beta}} \in T(S)$.

Definition 1.8.1. For every $(\alpha, \beta) \in \Delta \times \Delta$, put

$$t_{\alpha\beta} = (w_\alpha w_\beta)^{n_{\alpha\beta}} \in T(S).$$

Moreover, put (Exp. XX, 3.1)

$$t_\alpha = t_{\alpha\alpha} = w_\alpha^2 = \alpha^*(-1) \in T(S).$$

Proposition 1.8.2. Let H be an S-group functor taking direct sums of schemes to products (for example a sheaf for the Zariski topology). Let

$$f_T : T \rightarrow H$$

be a group morphism and h_α ($\alpha \in \Delta$) elements of $H(S)$. For there to exist a group morphism (necessarily unique)

$$f_N : \text{Norm}_G(T) \rightarrow H$$

which induces f_T on T and such that $f(w_\alpha) = h_\alpha$ for $\alpha \in \Delta$, it is necessary and sufficient that the following two conditions hold:

(i) For every $\alpha \in \Delta$, one has

$$f_T(s_\alpha(t)) = h_\alpha f_T(t) h_\alpha^{-1}$$

for every $t \in T(S')$, $S' \rightarrow S$ (i.e., $f_T \circ s_\alpha = \text{int}(h_\alpha) \circ f_T$).

(ii) For every $(\alpha, \beta) \in \Delta \times \Delta$, one has

$$f_T(t_{\alpha\beta}) = (h_\alpha h_\beta)^{n_{\alpha\beta}}.$$

⁽⁴⁾**N.D.E.** I.e., the Borel subgroup of Z_{Δ_1} corresponding to $R'_+ = R' \cap R_+$.

PROOF. Indeed, equip (Sch) with the topology $\mathcal{T}$ generated by the pretopology whose covering families are the direct sums; the hypothesis of the statement says that H is a $\mathcal{T}$ -sheaf. Let L be the free group with generators $(m_\alpha)_{\alpha \in R}$ and L_1 the invariant subgroup generated by the elements $(m_\alpha m_\beta)^{n_{\alpha\beta}}$, $(\alpha, \beta) \in \Delta \times \Delta$. Let $g : L \rightarrow W$ be the morphism defined by $g(m_\alpha) = s_\alpha$; one knows (Exp. XXI, 5.1) that g induces an isomorphism $\bar{g}$ from L/L_1 to W . Let L act on T through g (or, equivalently, by $m_\alpha \cdot t = s_\alpha(t)$). Let L_S be the constant group defined by L ; consider the semidirect product $T \cdot L_S = N$ for the preceding action. One has a morphism of S -groups

$$h : T \cdot L_S = N \rightarrow \underline{\text{Norm}}_G(T)$$

unique such that $h(m_\alpha) = w_\alpha$, $h(t) = t$ for every $t \in T(S')$, $S' \rightarrow S$. Let N_1 be the invariant group subsheaf of N generated by the

$$t_{\alpha\beta}^{-1} \cdot (m_\alpha m_\beta)^{n_{\alpha\beta}}, \quad (\alpha, \beta) \in \Delta \times \Delta.$$

One obviously has $N_1 \subset \text{Ker } h$; consider the induced morphism

$$h_1 : N/N_1 \rightarrow \underline{\text{Norm}}_G(T).$$

Let us prove that h_1 is an *isomorphism*. Since h induces on T the canonical immersion, which is a monomorphism, the canonical morphism

$$T \rightarrow N/N_1$$

is also a monomorphism, hence induces an isomorphism from T to TN_1/N_1 . For the same reason h_1 induces an isomorphism from TN_1/N_1 to T ; to prove that h_1 is an isomorphism, it therefore suffices to check that the corresponding morphism

$$h_2 : N/TN_1 \rightarrow \underline{\text{Norm}}_G(T)/T$$

is an isomorphism. Now TN_1 is the invariant group $\mathcal{T}$ -subsheaf of N generated by T and the $(m_\alpha m_\beta)^{n_{\alpha\beta}}$, that is, the $\mathcal{T}$ -subsheaf generated by T and L_1 , that is, $T \cdot (L_1)_S$. The morphism h_2 thus identifies with the morphism

$$\bar{g} : L_S/(L_1)_S \rightarrow W_S$$

which is an isomorphism by construction.

The proof of 1.8.2 is now easy; the conditions are obviously necessary; let us prove that they are sufficient. Condition (i) shows that there exists a morphism

$$u : N \rightarrow H$$

such that $u(m_\alpha) = h_\alpha$ for $\alpha \in \Delta$, and $u|_T = f_T$. Condition (ii) says that u vanishes on N_1 , which immediately gives the result. $\square$

1.9. — Faithfulness of the functor $\mathcal{R}$

Proposition 1.9.1. *The functor $\mathcal{R}$ of 1.6 is faithful: if*

$$f, g : G \rightrightarrows G'$$

are two morphisms of pinned groups such that $\mathcal{R}(f) = \mathcal{R}(g)$, then $f = g$.

Indeed, f and g coincide on T , U_α ($\alpha \in \Delta$) and $U_{-\alpha}$ ($\alpha \in \Delta$); it therefore suffices to apply:

Lemma 1.9.2. *Let S be a scheme, G a pinned reductive S -group, and H an S -group presheaf, separated for fppf. Let*

$$f, g : G \rightrightarrows H$$

be two morphisms of S -groups. The following conditions are equivalent:

- (i) $f = g$.
- (ii) f and g coincide on T , on each U_α ($\alpha \in \Delta$), on each $U_{-\alpha}$ ($\alpha \in \Delta$).
- (iii) f and g coincide on T , on each U_α ($\alpha \in \Delta$) and $f(w_\alpha) = g(w_\alpha)$ for each $\alpha \in \Delta$.

PROOF. Indeed, (i) $\Rightarrow$ (ii) is trivial, and (ii) $\Rightarrow$ (iii) follows immediately from the definition of w_α (1.2). It remains to prove (iii) $\Rightarrow$ (i). If $\alpha \in R$, there exists a sequence $\{\alpha_i\} \subset \Delta$ with $\alpha = s_{\alpha_1} s_{\alpha_2} \cdots s_{\alpha_n}(\alpha_{n+1})$, hence

$$U_\alpha = \text{int}(w_{\alpha_1} \cdots w_{\alpha_n}) U_{\alpha_{n+1}},$$

which proves that f and g coincide on each U_α . It follows that f and g coincide on Ω , hence coincide (Exp. XXII, 4.1.11). $\square$

Remark 1.9.3. If G is semisimple, one may, in (ii) and (iii), omit the hypothesis that f and g coincide on T . Indeed, G is generated as an fppf sheaf by the U_α , $\alpha \in R$ (Exp. XXII, 6.2.2 (a)).

2. Generators and relations for a pinned group

In this section, we fix a pinned S -group G . If $\alpha, \beta \in \Delta$, we shall use the notation Z_α , $Z_{\alpha\beta}$, $U_{\alpha\beta}$, $R_{\alpha\beta}^+$ of 1.7.

Theorem 2.1. *Let S be a scheme, G a pinned S -group, and H an S -group sheaf for fppf. Let*

$$f_N : \underline{\text{Norm}}_G(T) \longrightarrow H, \quad f_\alpha : U_\alpha \longrightarrow H, \quad \alpha \in R,$$

be group morphisms. For there to exist a group morphism (necessarily unique)

$$\bar{f} : G \longrightarrow H$$

extending f_N and the f_α ($\alpha \in R$), it is necessary and sufficient that the following conditions hold:

(i) *For every $\alpha \in \Delta$ and every $\beta \in R$, one has*

$$\text{int}(f_N(w_\alpha)) \circ f_\beta = f_{s_\alpha(\beta)} \circ \text{int}(w_\alpha).$$

(ii) *For every $\alpha \in \Delta$, there exists a group morphism*

$$F_\alpha : Z_\alpha \longrightarrow H$$

extending f_α , $f_{-\alpha}$ and $f_N|_{\underline{\text{Norm}}_{Z_\alpha}(T)}$.

(iii) *For every pair $(\alpha, \beta) \in \Delta \times \Delta$, there exists a group morphism $U_{\alpha\beta} \longrightarrow H$ inducing f_γ on U_γ for every $\gamma \in R_{\alpha\beta}^+$ (i.e., $U_\gamma \subset U_{\alpha\beta}$).*

2.1.1. — Proof. The conditions of the statement are obviously necessary. On the other hand, choose a totally ordered group structure on $\Gamma_0(R)$ so that the roots > 0 are the elements of R_+ (Exp. XXI, 3.5.6); every product indexed by a subset of R will be taken relative to this order. Denote by f_T the restriction of f_N to T and consider the morphism

$$f : \Omega \longrightarrow H$$

defined set-theoretically by

$$f \left(\prod_{\alpha \in R_-} y_\alpha \cdot t \cdot \prod_{\alpha \in R_+} x_\alpha \right) = \prod f_\alpha(y_\alpha) \cdot f_T(t) \cdot \prod f_\alpha(x_\alpha).$$

Every morphism satisfying the conditions of the statement must extend f ; on the other hand, every group morphism $\bar{f} : G \longrightarrow H$ extending f also extends f_N : indeed, extending f_α and $f_{-\alpha}$, it satisfies $\bar{f}(w_\alpha) = F_\alpha(w_\alpha) = f_N(w_\alpha)$ and it extends f_T by hypothesis. By Exp. XXII, 4.1.11 (ii), one is therefore reduced to proving:

Proposition 2.1.2. *The morphism $f : \Omega \longrightarrow H$ defined above is “generically multiplicative”; more precisely, for every $S' \longrightarrow S$ and all $x, y \in \Omega(S')$ such that $xy \in \Omega(S')$, one has $f(xy) = f(x)f(y)$.*

Lemma 2.1.3. *Let $n \in \underline{\text{Norm}}_G(T)(S')$, $S' \longrightarrow S$, and let $\alpha, \beta \in R$ be such that $\text{int}(n)U_\alpha = U_\beta$ (i.e., $\bar{n}(\alpha) = \beta$); then one has*

$$\text{int}(f_N(n)) \circ f_\alpha = f_\beta \circ \text{int}(n).$$

PROOF. Indeed, it suffices to check the formula for a system of generators of the sheaf $\underline{\text{Norm}}_G(T)$; it holds for each w_α , $\alpha \in \Delta$ (by 2.1 (i)), so it suffices to do so for $n \in T(S')$. This is trivial by 2.1 (ii) if α is simple; otherwise, one takes a $w \in W$ such that $w^{-1}(\alpha) \in \Delta$; writing w as a product of simple reflections,⁽⁵⁾ one is reduced to proving that if the formula holds for α and for every n , it also holds for $w_{\alpha_0}(\alpha)$ and $t \in T(S')$, where $\alpha_0 \in \Delta$. Now, by 2.1 (i), one has:

$$\begin{aligned} \text{int}(f_N(t)) \circ f_{w_{\alpha_0}(\alpha)} &= \text{int}(f_N(tw_{\alpha_0})) \circ f_\alpha \circ \text{int}(w_{\alpha_0}^{-1}) \\ &= f_{w_{\alpha_0}(\alpha)} \circ \text{int}(tw_{\alpha_0}) \circ \text{int}(w_{\alpha_0}^{-1}). \end{aligned}$$

□

Lemma 2.1.4. *The restriction of f to U (resp. U^-) is a group morphism. In particular, this restriction is independent of the order chosen on the roots.*

PROOF. It suffices to give the proof for U . By Exp. XXII, 5.5.8, it suffices to check that for every pair $\alpha < \beta$ of positive roots, one has, for all $x_\alpha \in U_\alpha(S')$, $x_\beta \in U_\beta(S')$, $S' \longrightarrow S$,

$$f_\beta(x_\beta)f_\alpha(x_\alpha)f_\beta(x_\beta^{-1}) = f(x_\beta x_\alpha x_\beta^{-1}).$$

According to Exp. XXII, 5.5.2, there exist $x_\gamma \in U_\gamma(S')$ ($\gamma = i\alpha + j\beta \in R$, $i > 0$, $j \geq 0$)⁽⁶⁾ such that

$$x_\beta x_\alpha x_\beta^{-1} = \prod_{\gamma} x_\gamma,$$

and one must therefore check the relation

$$f_\beta(x_\beta)f_\alpha(x_\alpha)f_\beta(x_\beta^{-1}) = \prod_{\substack{\gamma=i\alpha+j\beta \\ i>0, j\geq 0}} f_\gamma(x_\gamma).$$

By Exp. XXI, 3.5.4, there exists a $w \in W$ such that $w(\alpha) = \alpha_0 \in \Delta$, $w(\beta) \in A_{\alpha_0\beta_0}$ (notation of 1.7), where $\beta_0 \in \Delta$. Lifting w to an $n \in \underline{\text{Norm}}_G(T)(S)$ (by Exp. XXII, 3.8), it suffices to check the preceding relation after conjugation by $f_N(n)$. By 2.1.3, one is therefore reduced to the case where $\alpha, \beta \in R_{\alpha_0\beta_0}^+$, a case where one concludes by condition 2.1 (iii). □

Lemma 2.1.5. *Let $n \in \underline{\text{Norm}}_G(T)(S)$ be such that $\text{int}(n)U = U^-$ (i.e., $\bar{n}$ is the symmetry of the Weyl group⁽⁷⁾ (Exp. XXI, 3.6.14)). For every $u \in U(S')$, $S' \longrightarrow S$ (resp. $u \in U^-(S')$, $S' \longrightarrow S$), one has*

$$f(nun^{-1}) = f_N(n)f(u)f_N(n^{-1}).$$

PROOF. Immediate by 2.1.3 and 2.1.4. □

Lemma 2.1.6. *Let $u \in B(S')$, $v \in B^-(S')$, $g \in \Omega(S')$, $S' \longrightarrow S$. Then*

$$f(vgu) = f(v)f(g)f(u).$$

PROOF. Indeed, put $v = v_1t_1$, $g = v_2t_2u_2$, $u = t_3u_3$, with $v_i \in U^-(S')$, $t_i \in T(S')$, $u_i \in U(S')$. One has

$$f(v)f(g)f(u) = f(v_1)f_T(t_1)f(v_2)f_T(t_2)f(u_2)f_T(t_3)f(u_3),$$

⁽⁵⁾**N.D.E.** We have replaced “fundamental symmetries” by “simple reflections”.

⁽⁶⁾**N.D.E.** We have corrected $i, j > 0$ to $i > 0$, $j \geq 0$ (since x_α occurs in the product on the right).

⁽⁷⁾**N.D.E.** With respect to Δ .

on the one hand, and

$$\begin{aligned} f(vgu) &= f(v_1 t_1 v_2 t_1^{-1} t_1 t_2 t_3 t_3^{-1} u_2 t_3 u_3) \\ &= f(v_1 \cdot t_1 v_2 t_1^{-1}) f_T(t_1 t_2 t_3) f(t_3^{-1} u_2 t_3 \cdot u_3). \end{aligned}$$

Using 2.1.4 to decompose the two outer terms of this last expression, one obtains

$$f(vgu) = f(v_1) f(t_1 v_2 t_1^{-1}) f_T(t_1 t_2 t_3) f(t_3^{-1} u_2 t_3) f(u_3).$$

One is then reduced to the obvious formulas

$$\begin{aligned} f(t_1 v_2 t_1^{-1}) &= f_T(t_1) f(v_2) f_T(t_1)^{-1}, \\ f(t_3^{-1} u_2 t_3) &= f_T(t_3)^{-1} f(u_2) f_T(t_3), \end{aligned}$$

which follow from the definition of f and from 2.1.3. $\square$

Lemma 2.1.7. *Let $\alpha \in \Delta$ and $g \in \Omega(S') \cap \text{int}(w_\alpha)^{-1} \Omega(S')$, $S' \rightarrow S$. Then*

$$f(w_\alpha g w_\alpha^{-1}) = f_N(w_\alpha) f(g) f_N(w_\alpha)^{-1}.$$

PROOF. Indeed, let $g \in \Omega(S')$, $S' \rightarrow S$. Write, by Exp. XXII, 5.6.8,

$$g = a x_{-\alpha} t x_\alpha b,$$

with $a \in U_{-\hat{\alpha}}(S')$, $x_{-\alpha} \in U_{-\alpha}(S')$, $t \in T(S')$, $x_\alpha \in U_\alpha(S')$, $b \in U_{\hat{\alpha}}(S')$. By 2.1.3, 2.1.4 and the fact that s_α permutes the positive roots $\neq \alpha$ (cf. Exp. XXI, 3.3.2), one has

$$\text{int}(w_\alpha) a \in U_{-\hat{\alpha}}(S'), \quad \text{int}(w_\alpha) b \in U_{\hat{\alpha}}(S')$$

and the formula to be proved holds for $g = a$ or $g = b$. By 2.1.6, one is therefore reduced to proving the desired assertion when $g = x_{-\alpha} t x_\alpha \in Z_\alpha(S')$. But then “everything takes place in Z_α ” and one concludes by condition (ii) of 2.1. $\square$

Lemma 2.1.8. *Let $n \in \underline{\text{Norm}}_G(T)(S)$. For every $S' \rightarrow S$ and every $g \in \Omega(S')$ such that $\text{int}(n)g \in \Omega(S')$, one has*

$$f(n g n^{-1}) = f_N(n) f(g) f_N(n)^{-1}.$$

PROOF. This is trivial if $n \in T(S)$ (by 2.1.3). The two sides of the preceding formula define morphisms from $\Omega \cap \text{int}(n)^{-1} \Omega$ to H ; to check that they coincide, it suffices to check that there exists an open subset V_n of Ω , containing the identity section, such that $\text{int}(n)V_n \subset \Omega$, and that $f \circ \text{int}(n)$ and $\text{int}(f_N(n)) \circ f$ coincide on V_n . By the structure of $\underline{\text{Norm}}_G(T)$, it suffices to prove that if the preceding assertion holds for an $n' \in \underline{\text{Norm}}_G(T)(S)$ and if $\alpha \in \Delta$, it holds for $n = n' w_\alpha$. Put

$$V_n = \Omega \cap \text{int}(w_\alpha)^{-1} V_{n'}.$$

One has $\text{int}(n)V_n \subset \text{int}(n')V_{n'} \subset \Omega$. If $g \in V_n(S')$, one has

$$\text{int}(n)g = \text{int}(n') \text{int}(w_\alpha)g.$$

Now $\text{int}(w_\alpha)g \in V_{n'}(S')$, so by hypothesis

$$f(\text{int}(n') \text{int}(w_\alpha)g) = \text{int}(f_N(n')) f(\text{int}(w_\alpha)g);$$

since $\text{int}(w_\alpha)g \in \Omega(S')$, one may apply 2.1.7, which gives

$$f(\text{int}(w_\alpha)g) = \text{int}(f_N(w_\alpha)) f(g),$$

and one concludes immediately. $\square$

Let us now prove 2.1.2. Let $x, x' \in \Omega(S')$; write as usual

$$x = vtu, \quad x' = v't'u',$$

whence

$$xx' = vt(uv')t'u'.$$

By 2.1.6 and the relation $U^-(S')\Omega(S')U(S') = \Omega(S')$, one is reduced to proving that if $uv' \in \Omega(S')$, one has $f(uv') = f(u)f(v')$. Let $n \in \underline{\text{Norm}}_G(T)(S')$ be as in 2.1.5 (ii). One has

$$\begin{aligned} f(u) &= f_N(n)^{-1}f(nun^{-1})f_N(n), \\ f(v') &= f_N(n)^{-1}f(nv'n^{-1})f_N(n), \end{aligned}$$

by *loc. cit.*, whence

$$f(u)f(v') = f_N(n)^{-1}f(nun^{-1})f(nv'n^{-1})f_N(n).$$

But $nun^{-1} \in U^-(S')$, $nv'n^{-1} \in U(S')$, so that

$$f(nun^{-1})f(nv'n^{-1}) = f((nun^{-1})(nv'n^{-1})) = f(nuv'n^{-1}),$$

which gives

$$f(u)f(v') = f_N(n)^{-1}f(nuv'n^{-1})f_N(n).$$

If $uv' \in \Omega(S')$, one may apply 2.1.8 and the proof is complete.

Remark 2.2. Instead of giving f_N , one may give a group morphism $f_T : T \rightarrow H$, and sections $h_\alpha \in H(S)$ ($\alpha \in \Delta$) satisfying the conditions of 1.8.2. One must then replace condition (ii) of the theorem by

(ii') There exists a group morphism $F_\alpha : Z_\alpha \rightarrow H$ which extends

$$f_\alpha, f_{-\alpha}, f_T \quad \text{and satisfies } F_\alpha(w_\alpha) = h_\alpha.$$

We shall now give the preceding theorem a more explicit form. Retain the preceding notation.

Theorem 2.3. *Let S be a scheme, and G a pinned S -group. Let H be an S -group sheaf for fppf,*

$$f_T : T \rightarrow H, \quad f_\alpha : U_\alpha \rightarrow H, \quad \alpha \in \Delta,$$

group morphisms, and

$$h_\alpha \in H(S), \quad (\alpha \in \Delta),$$

sections of H . For there to exist a group morphism

$$f : G \rightarrow H$$

(necessarily unique) inducing f_T and the f_α ($\alpha \in \Delta$) and satisfying $f(w_\alpha) = h_\alpha$ for every $\alpha \in \Delta$, it is necessary and sufficient that the following conditions hold:

- (i) *For every $S' \rightarrow S$, every $t \in T(S')$, every $\alpha \in \Delta$ and every $x \in U_\alpha(S')$, one has*

$$(1) \quad f_T(t)f_\alpha(x)f_T(t)^{-1} = f_\alpha(\text{int}(t)x).$$
- (ii) *For every $\alpha \in \Delta$, every $S' \rightarrow S$, every $t \in T(S')$, one has*

$$(2) \quad h_\alpha f_T(t)h_\alpha^{-1} = f_T(s_\alpha(t)) = f_T(t \cdot \alpha^* \alpha(t)^{-1}).$$
- (iii) *For every $(\alpha, \beta) \in \Delta \times \Delta$, one has*

$$(3) \quad (h_\alpha h_\beta)^{n_{\alpha\beta}} = f_T(t_{\alpha\beta}).$$
- (iv) *For every $\alpha \in \Delta$, one has (recall that one writes $u_\alpha = \exp_\alpha(X_\alpha)$)*

$$(4) \quad (h_\alpha f_\alpha(u_\alpha))^3 = e.$$
- (v) *For every $(\alpha, \beta) \in \Delta \times \Delta$, $\alpha \neq \beta$, there exists a group morphism*

$$f_{\alpha\beta} : U_{\alpha\beta} \rightarrow H$$

satisfying the following two conditions:

a) One has

$$(5) \quad f_{\alpha\beta}|_{U_\alpha} = f_\alpha, \quad f_{\alpha\beta}|_{U_\beta} = f_\beta,$$

b) For every $\gamma \in R_{\alpha\beta}^+$, $\gamma \neq \alpha$ (resp. $\gamma \neq \beta$), and every $x \in U_\gamma(S')$, $S' \rightarrow S$, one has

$$(6) \quad \begin{aligned} \text{int}(h_\alpha)f_{\alpha\beta}(x) &= f_{\alpha\beta}(\text{int}(w_\alpha)x), \\ (\text{resp. } \text{int}(h_\beta)f_{\alpha\beta}(x) &= f_{\alpha\beta}(\text{int}(w_\alpha)x)). \end{aligned}$$

Proof. Uniqueness is clear by 1.9.2. Let us prove existence.

Lemma 2.3.1. *There exists a group morphism*

$$f_N : \underline{\text{Norm}}_G(T) \rightarrow H$$

extending f_T and satisfying $f_N(w_\alpha) = h_\alpha$.

Indeed, this is what 1.8.2 asserts, taking conditions (2) and (3) into account.

Lemma 2.3.2. *There exists a group morphism*

$$F_\alpha : Z_\alpha \rightarrow H, \quad \alpha \in \Delta,$$

extending f_T , f_α and satisfying $F_\alpha(w_\alpha) = h_\alpha$, hence extending $f_N|_{\underline{\text{Norm}}_{Z_\alpha}(T)}$.

This is clear by Exp. XX, 6.2 and conditions (1), (3) and (4).

Lemma 2.3.3. *For every $(\alpha, \beta) \in \Delta \times \Delta$, $\alpha \neq \beta$, every $n \in \underline{\text{Norm}}_{Z_{\alpha\beta}}(T)(S)$ such that $\bar{n}(\alpha) = \alpha$ (resp. $\bar{n}(\alpha) = \beta$), i.e., $\text{int}(n)U_\alpha = U_\alpha$ (resp. $\text{int}(n)U_\alpha = U_\beta$), every $S' \rightarrow S$ and every $x \in U_\alpha(S')$, one has*

$$\begin{aligned} \text{int}(f_N(n))f_\alpha(x) &= f_\alpha(\text{int}(n)x) \\ \text{resp. } \text{int}(f_N(n))f_\alpha(x) &= f_\beta(\text{int}(n)x). \end{aligned}$$

PROOF. Indeed, there exist a $t \in T(S)$ and a sequence $\{\alpha_i\} \subset \{\alpha, \beta\}$ such that $n = tw_{\alpha_k} \cdots w_{\alpha_1}$. The condition holds for $n = t$ by condition (1). One may therefore suppose $n = w_{\alpha_k} \cdots w_{\alpha_1}$. Let us give the proof by induction on k , i.e., suppose the assertion proved for every n which can be written as a product of fewer than $k - 1$ simple reflections (and satisfies the hypotheses of the lemma). Consider the roots

$$\gamma_i = s_{\alpha_i} \cdots s_{\alpha_1}(\alpha).$$

If all the γ_i are positive, i.e., belong to $R_{\alpha\beta}^+$, one may apply condition (v) of 2.3; adopting the notation of 2.3 (v), one concludes immediately by induction that

$$\text{int}(f_N(w_{\alpha_i} \cdots w_{\alpha_1}))f_\alpha(x) = f_{\alpha\beta}(\text{int}(w_{\alpha_i} \cdots w_{\alpha_1})x),$$

which, for $i = k$, is the desired property. If not all the γ_i are positive, there exists an index $j < k$ such that

$$\gamma_j \in R_+, \quad \gamma_{j+1} \notin R_+.$$

Since $\gamma_{j+1} = s_{\alpha_j}(\gamma_j)$, it follows immediately that $\gamma_j = \alpha_j$, by Exp. XXI, 3.3.2, hence that γ_j is α or β , and one may decompose n as

$$\begin{aligned} n &= n' \cdot n'', \\ n' &= w_{\alpha_k} \cdots w_{\alpha_{j+1}}, \\ n'' &= w_{\alpha_j} \cdots w_{\alpha_1}, \end{aligned}$$

n' and n'' satisfying the hypotheses of the lemma and being products of fewer than $k - 1$ reflections, hence satisfying the conclusion of the lemma by the induction hypothesis. $\square$

Lemma 2.3.4. *Let $\alpha \in R$. If $n, n' \in \underline{\text{Norm}}_G(T)(S)$ and $\beta, \beta' \in \Delta$ satisfy $\bar{n}(\alpha) = \beta$ and $\bar{n}'(\alpha) = \beta'$, one has*

$$\text{int}(f_T(n)^{-1})f_\beta(nxn^{-1}) = \text{int}(f_T(n')^{-1})f_{\beta'}(n'xn'^{-1}),$$

for every $x \in U_\alpha(S')$, $S' \rightarrow S$.

PROOF. It suffices to check that if $\bar{n}(\alpha) = \beta$, $\alpha, \beta \in \Delta$, then $\text{int}(f_T(n)) \circ f_\alpha = f_\beta \circ \text{int}(n)$. Now, by Tits's lemma (Exp. XXI, 5.6), there exist a sequence of simple roots $\alpha_0 = \alpha$, $\alpha_1, \dots, \alpha_m = \beta$, and a sequence of elements $w_i \in W$, $i = 0, 1, \dots, m-1$, with

$$\begin{aligned} \bar{n} &= w_{m-1} \cdots w_0, \\ w_i(\alpha_i) &= \alpha_{i+1}, \quad i = 0, 1, \dots, m-1, \end{aligned}$$

the following condition being moreover satisfied: for every $i = 0, 1, \dots, m-1$, there exists a simple root β_i such that

$$w_i \in W_{\alpha_i \beta_i}, \quad \alpha_{i+1} = \alpha_i \quad \text{or} \quad \beta_i.$$

One is then reduced by induction to the case treated in the preceding lemma. $\square$

Lemma 2.3.5. *There exists a family of group morphisms $f'_\gamma : U_\gamma \rightarrow H$, $\gamma \in R$, satisfying*

- (i) *For $\alpha \in \Delta$, one has $f'_\alpha = f_\alpha$ and $f'_{-\alpha} = F_\alpha|_{U_{-\alpha}}$, where F_α is defined in 2.3.2.*
- (ii) *For $\alpha, \beta \in \Delta$ and $\gamma \in R_{\alpha\beta}^+$, one has $f'_\gamma = f_{\alpha\beta}|_{U_\gamma}$.*
- (iii) *For every $n \in \text{Norm}_G(T)(S)$ and $\alpha, \beta \in R$ such that $\bar{n}(\alpha) = \beta$, one has*

$$\text{int}(f_N(n))f'_\alpha(x) = f'_\beta(\text{int}(n)x)$$

for every $x \in U_\alpha(S')$, $S' \rightarrow S$.

PROOF. For every root $\alpha \in R$, there exists an $n \in \text{Norm}_G(T)(S)$ such that $\bar{n}(\alpha) \in \Delta$. Define $f'_\alpha(x)$ to be the expression of 2.3.4. This is independent of the choice of n and f'_α is indeed a group morphism. Property (iii) holds by construction. The first assertion of (i) is clear (take $n = e$), and so is the second (take $n = w_\alpha$ and apply 2.3.2); if $\gamma \in R_{\alpha\beta}^+$, $(\alpha, \beta \in \Delta)$, there exists $n \in \text{Norm}_{Z_{\alpha\beta}}(T)(S)$ such that $\bar{n}(t) = \alpha$ or β ; one then applies (iii) and conditions (5) and (6) and has proved (ii). $\square$

2.3.6. — Let us now prove the theorem by proving that the conditions of 2.1 hold for the morphisms f_N and f'_α , $\alpha \in R$.

- 2.1 (i) follows from 2.3.5 (iii),
 - 2.1 (ii) follows from 2.3.5 (i) and 2.3.2,
 - 2.1 (iii) follows from 2.3.5 (ii) and the fact that $f_{\alpha\beta}$ is a group morphism.
- An obvious corollary is:

Corollary 2.4. *Let S be a scheme, G a pinned S -group of semisimple rank ≥ 1 , and H an fppf S -group sheaf. For each $(\alpha, \beta) \in \Delta \times \Delta$, let*

$$F_{\alpha\beta} : Z_{\alpha\beta} \rightarrow H$$

be a group morphism. For there to exist a group morphism

$$f : G \rightarrow H$$

inducing the $F_{\alpha\beta}$, it is necessary and sufficient that for every $(\alpha, \beta) \in \Delta \times \Delta$, one have

$$F_{\alpha\beta}|_{Z_\alpha} = F_{\alpha\alpha}.$$

PROOF. The condition is obviously necessary. Suppose it satisfied. Put $f_T = F_{\alpha\alpha}|_T$ (which does not depend on α , since $F_{\alpha\alpha}|_T = F_{\alpha\beta}|_T = F_{\beta\beta}|_T$). Put

$$p_\alpha = F_{\alpha\alpha}|_{U_\alpha}, \quad h_\alpha = F_{\alpha\alpha}(w_\alpha), \quad f_{\alpha\beta} = F_{\alpha\beta}|_{U_{\alpha\beta}}.$$

The conditions of 2.3 are obviously satisfied: (1), (2), (4) “are checked” in Z_α , (3), (5) and (6) in $Z_{\alpha\beta}$. There thus exists a morphism $f : G \rightarrow H$ extending f_T , the f_α , $\alpha \in \Delta$, and satisfying $f(w_\alpha) = h_\alpha$; it coincides with $F_{\alpha\beta}$ on generators of $Z_{\alpha\beta}$, hence satisfies $f|_{Z_{\alpha\beta}} = F_{\alpha\beta}$. $\square$

One also has the following technical corollary:

Corollary 2.5. *Let S be a scheme, G and G' two pinned S -groups of semisimple rank 2, q an integer > 0 such that $x \mapsto x^q$ is an endomorphism of $\mathbf{G}_{a,S}$, and $h : \mathcal{R}(G') \rightarrow \mathcal{R}(G)$*

a q -morphism of pinned root data. For each $\alpha \in R_+$, choose an $X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times$ and an $X'_{d(\alpha)} \in \Gamma(S, \mathfrak{g}^{d(\alpha)})^\times$ (extending the canonical choices for $\alpha \in \Delta$). Suppose the following conditions hold:

- (i) If $\Delta = \{\alpha, \beta\}$, one has $D_S(h)(t_{\alpha\beta}) = t'_{d(\alpha)d(\beta)}$.
- (ii) For every $\alpha \in \Delta$ and every $\beta \in R_+$, $\beta \neq \alpha$ (whence $s_\alpha(\beta) \in R_+$), if $z \in \mathbf{G}_m(S)$ is defined by

$$\mathrm{Ad}(w_\alpha)X_\beta = zX_{s_\alpha(\beta)},$$

one also has

$$\mathrm{Ad}(w'_{d(\alpha)})X'_{d(\beta)} = z^{q(\beta)}X'_{d(s_\alpha(\beta))}.$$

- (iii) There exists a group morphism $f : U \rightarrow U'$ such that for every $\alpha \in R_+$ one has, for every $x \in \mathbf{G}_a(S')$, $S' \rightarrow S$,

$$f(\exp(xX_\alpha)) = \exp(x^{q(\alpha)}X'_{d(\alpha)}).$$

Then there exists a morphism of pinned groups $G \xrightarrow{g} G'$ such that $\mathcal{R}(g) = h$.

PROOF. Indeed, one defines $f_\alpha : U_\alpha \rightarrow G'$ by

$$f_\alpha(\exp(xX_\alpha)) = \exp(x^{q(\alpha)}X'_{d(\alpha)});$$

one puts $f_T = D_S(h)$, $h_\alpha = w'_{d(\alpha)}$. The conditions of 2.3 are satisfied (observe that $q(s_\alpha(\beta)) = q(\beta)$ (Exp. XXI, 6.8.4) and that one always has $D_S(h)(t_\alpha) = t'_{d(\alpha)}$) and one concludes immediately. $\square$

Remark 2.6. One may make condition (v) of 2.3 more precise as follows. One must first check:

(a) For every word in w_α and w_β such that the corresponding word takes α or β to α or β , the corresponding relation of type 2.3.3 holds. In fact the proof of 2.3.3 shows that it suffices to check it for the words in w_α and w_β which are minimal (in the sense that every nontrivial initial subword fails to satisfy the imposed conditions).

If condition (a) holds, one may now define, for each $\gamma \in R_{\alpha\beta}^+$, an $f_\gamma : U_\gamma \rightarrow H$ as in 2.3.5; one must then write:

(b) The morphism defined by the f_γ

$$U_{\alpha\beta} = \prod_{\gamma \in R_{\alpha\beta}^+} U_\gamma \rightarrow H$$

is a group morphism. According to Exp. XXII, 5.5.8, (b) is implied by:

(b') The preceding morphism respects the commutation relations between U_γ and U_δ for $\gamma, \delta \in R_{\alpha\beta}^+$, $\gamma > \delta$ (i.e., the relations in $C_{i,j,\gamma,\delta}$ of Exp. XII, 5.5.2).

It is clear that, conversely, the collection of conditions (a) and (b') is equivalent to (v).

One may even reduce the preceding conditions to conditions involving only the elements h_α , h_β , $f_\alpha(u_\alpha)$, $f_\beta(u_\beta)$ of H . A condition of type (a) will be written, for example, if $s_\alpha s_\beta s_\alpha(\beta) = \alpha$:

$$(1) \quad \mathrm{int}(h_\alpha h_\beta h_\alpha) f_\beta(x) = f_\alpha(\mathrm{int}(w_\alpha w_\beta w_\alpha)x);$$

for every $x \in U_\alpha(S')$, $S' \rightarrow S$. In particular, for $x = u_\beta$, one has $\mathrm{int}(w_\alpha w_\beta w_\alpha)u_\beta = u_\alpha^z$ for a certain section z of $\mathbf{G}_m(S)$, and the preceding relation gives

$$(1') \quad \mathrm{int}(h_\alpha h_\beta h_\alpha) f_\beta(u_\beta) = f_\alpha(u_\alpha)^z.$$

Let us show that, conversely, taking conditions (i) to (iv) of 2.3 into account, (1') implies (1). If $t \in T(S')$, $S' \rightarrow S$, let $\mathrm{int}(t)$ act on (1'); taking conditions (i) and (ii) into account, one obtains (1) for $x = \mathrm{int}(t)u_\beta = u_\beta^{\beta(t)}$. It now suffices to observe that $\beta : T \rightarrow \mathbf{G}_{m,S}$ is faithfully flat, hence that condition (1) certainly holds for $x \in U_\alpha(S')^\times$, $S' \rightarrow S$. Since it is additive in x and every section of U_α can locally be written as a sum of two sections of $U_\alpha^\times$, one does indeed conclude that (1') + (i) + (ii) $\implies$ (1).

One argues in the same way with conditions of type (b). One must then use the fact that if γ and γ' are two distinct positive roots (and hence linearly independent over $\mathbf{Q}$), the morphism $T \rightarrow \mathbf{G}_{m,S}^2$ with components γ and γ' is faithfully flat. We leave the details of this transposition to the reader.

3. Groups of semisimple rank 2

3.1. — Generalities

Lemma 3.1.1. *Let S be a scheme, G a pinned S -group, and α and β two roots of G , with $\alpha + \beta \neq 0$.*

(i) *If $\alpha + \beta \notin R$, one has*

$$\exp(X_\alpha) \exp(X_\beta) = \exp(X_\beta) \exp(X_\alpha)$$

for all $X_\alpha \in \mathbf{W}(\mathfrak{g}^\alpha)(S')$, $X_\beta \in \mathbf{W}(\mathfrak{g}^\beta)(S')$, $S' \rightarrow S$.

(ii) *If $\alpha + \beta$ and $\alpha - \beta$ are not roots, one has*

$$w_\alpha(z_\alpha) \exp(X_\beta) w_\alpha(z_\alpha)^{-1} = \exp(X_\beta)$$

for all $X_\beta \in \mathbf{W}(\mathfrak{g}^\beta)(S')$, $z_\alpha \in \mathbf{W}(\mathfrak{g}^\alpha)^\times(S')$, $S' \rightarrow S$, and

$$w_\alpha(z_\alpha) w_\beta(z_\beta) = w_\beta(z_\beta) w_\alpha(z_\alpha)$$

for all $z_\alpha \in \mathbf{W}(\mathfrak{g}^\alpha)^\times(S')$ and $z_\beta \in \mathbf{W}(\mathfrak{g}^\beta)^\times(S')$, $S' \rightarrow S$.

(iii) *Let $X_\alpha \in \mathbf{W}(\mathfrak{g}^\alpha)^\times(S')$, $X_\beta \in \mathbf{W}(\mathfrak{g}^\beta)^\times(S')$, and $w \in \underline{\text{Norm}}_G(T)(S')$ be such that $\bar{w}(\alpha) = \beta$; define $z \in \mathbf{G}_m(S')$ by*

$$\text{Ad}(w)X_\alpha = zX_\beta.$$

Then

$$\text{int}(w) \exp(xX_\alpha) = \exp(xzX_\beta),$$

$$\text{int}(w) \exp(yX_\alpha^{-1}) = \exp(yz^{-1}X_\beta^{-1}),$$

$$\text{int}(w)w_\alpha(X_\alpha) = \beta^*(z)w_\beta(X_\beta).$$

In particular, if $z = \pm 1$, one has

$$\text{int}(w)w_\alpha(X_\alpha) = w_\beta(X_\beta)^z.$$

(iv) *If one puts $t_\alpha = \alpha^*(-1)$, $t_\beta = \beta^*(-1)$, one has*

$$s_\alpha(t_\beta) = t_\beta t_\alpha^{(\beta^*, \alpha)}, \quad \beta(t_\alpha) = (-1)^{(\alpha^*, \beta)}.$$

PROOF. (i) follows immediately from Exp. XXII, 5.5.2, (ii) from Exp. XX, 3.1 and from (i) applied to (β, α) , $(\beta, -\alpha)$, $(-\beta, \alpha)$, $(-\beta, -\alpha)$; (iii) is evident from the definitions; for the last assertion of (iii), observe that $\beta^*(-1) = w_\beta(X_\beta)^{-2}$. Finally, (iv) is trivial. $\square$

Proposition 3.1.2. (Groups of type $A_1 \times A_1$). *Let S be a scheme, G a pinned S -group of type $A_1 \times A_1$; write $\Delta = R_+ = \{\alpha, \beta\}$.*

(i) *One has*

$$t_{\alpha\beta} = (w_\alpha w_\beta)^2 = t_\alpha t_\beta = (w_\beta w_\alpha)^2 = t_{\beta\alpha}.$$

(ii) *One has*

$$\text{Ad}(w_\alpha)X_\beta = X_\beta, \quad \text{Ad}(w_\beta)X_\alpha = X_\alpha.$$

(iii) *U_α and U_β commute (i.e., U is commutative).*

PROOF. Indeed, by assertion (ii) of the lemma, one has

$$w_\alpha w_\beta = w_\beta w_\alpha,$$

whence $(w_\alpha w_\beta)^2 = w_\alpha^2 w_\beta^2 = t_\alpha t_\beta$, which is (i). By assertion (ii) of the lemma, one also has (ii); finally, (iii) is assertion (i) of the lemma. $\square$

3.1.3. — Let us make condition (v) of 2.3 explicit here. Using the method explained in 2.6, one obtains the following two groups of conditions, putting $v_\alpha = f_\alpha(u_\alpha)$, for $\alpha \in \Delta$:

$$(A) \quad \begin{cases} h_\alpha v_\beta h_\alpha^{-1} = v_\beta, \\ h_\beta v_\alpha h_\beta^{-1} = v_\alpha, \end{cases} \quad (B) \quad v_\alpha v_\beta = v_\beta v_\alpha.$$

3.2. — Groups of type A_2

Proposition 3.2.1. *Let S be a scheme, G a pinned S -group of type A_2 ; write $\Delta = \{\alpha, \beta\}$, $R_+ = \{\alpha, \beta, \alpha + \beta\}$.*

(i) *One has*

$$t_{\alpha\beta} = (w_\alpha w_\beta)^3 = e = (w_\beta w_\alpha)^3 = t_{\beta\alpha}.$$

(ii) *Put $\text{Ad}(w_\beta)X_\alpha = X_{\alpha+\beta}$. Then*

$$\begin{aligned} \text{Ad}(w_\alpha)X_\beta &= -X_{\alpha+\beta}, & \text{Ad}(w_\alpha)X_{\alpha+\beta} &= X_\beta, \\ \text{Ad}(w_\beta)X_{\alpha+\beta} &= -X_\alpha. \end{aligned}$$

(iii) *Put $p_{\alpha+\beta}(x) = \exp(xX_{\alpha+\beta}) = \text{int}(w_\beta)p_\alpha(x)$. One has:*

$$p_\beta(y)p_\alpha(x) = p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(xy).$$

3.2.2. — The proof occupies nos. 3.2.2 to 3.2.7. First, one has $(\beta^*, \alpha) = (\alpha^*, \beta) = -1$, whence $\alpha(t_\beta) = \beta(t_\alpha) = -1$.

Put $\text{Ad}(w_\beta)X_\alpha = X_{\alpha+\beta}$; one immediately has

$$\text{Ad}(w_\beta)X_{\alpha+\beta} = \alpha(t_\beta)X_\alpha = -X_\alpha.$$

Put

$$\text{Ad}(w_\alpha)X_\beta = zX_{\alpha+\beta}, \quad z \in \mathbf{G}_m(S),$$

whence

$$\text{Ad}(w_\alpha)X_{\alpha+\beta} = \beta(t_\alpha)z^{-1}X_\beta = -z^{-1}X_\beta.$$

We know (Exp. XXII, 5.5.2) that there exists a unique section $A \in \mathbf{G}_a(S)$ such that

$$(+) \quad p_\beta(y)p_\alpha(x) = p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(Axy).$$

Thus, to prove (ii) and (iii), the task is to show that $A = -z = 1$.

3.2.3. — Let $\text{int}(w_\beta)$ act on the preceding formula (+); one obtains:

$$(++) \quad p_{-\beta}(-y)p_{\alpha+\beta}(x) = p_{\alpha+\beta}(x)p_{-\beta}(-y)p_\alpha(-Axy).$$

3.2.4. — By definition of $p_{\alpha+\beta}$, one has

$$w_\beta p_\alpha(x) w_\beta^{-1} = p_{\alpha+\beta}(x),$$

which can be written

$$\begin{aligned} p_\beta(1)p_{-\beta}(-1)p_\beta(1)p_\alpha(x)p_\beta(-1)p_{-\beta}(1)p_\beta(-1) \\ = p_{\alpha+\beta}(x). \end{aligned}$$

Since p_β and $p_{\alpha+\beta}$ commute, $\alpha + 2\beta$ not being a root, this can also be written

$$p_\beta(1)p_\alpha(x)p_\beta(-1) = p_{-\beta}(1)p_{\alpha+\beta}(x)p_{-\beta}(-1).$$

Now using (+) on the left and (++) on the right, one obtains:

$$\begin{aligned} p_\alpha(x)p_\beta(1)p_{\alpha+\beta}(Ax)p_\beta(-1) \\ = p_{\alpha+\beta}(x)p_{-\beta}(1)p_\alpha(Ax)p_{-\beta}(-1). \end{aligned}$$

Since $\alpha + 2\beta$ (resp. $\alpha - \beta$) is not a root, the left (resp. right) side can be written

$$p_\alpha(x)p_{\alpha+\beta}(Ax) \quad \text{resp.} \quad p_{\alpha+\beta}(x)p_\alpha(Ax)$$

and the term on the right equals $p_\alpha(Ax)p_{\alpha+\beta}(x)$, since $2\alpha + \beta$ is not a root. Thus

$$p_\alpha(x)p_{\alpha+\beta}(Ax) = p_\alpha(Ax)p_{\alpha+\beta}(x)$$

which gives (by XXII 4.1.3) $A = 1$.

3.2.5. — Now let $\text{int}(w_\alpha)$ act on formula (+); one finds, using the fact that $A = 1$,

$$(+++) \quad p_{\alpha+\beta}(zy)p_{-\alpha}(-x) = p_{-\alpha}(-x)p_{\alpha+\beta}(zy)p_{\beta}(-z^{-1}xy).$$

3.2.6. — Let us now write, as before,

$$w_\alpha p_\beta(y)w_\alpha^{-1} = p_{\alpha+\beta}(zy),$$

whence, since p_α and $p_{\alpha+\beta}$ commute,

$$p_\alpha(1)p_\beta(y)p_\alpha(-1) = p_{-\alpha}(1)p_{\alpha+\beta}(zy)p_{-\alpha}(-1).$$

Now using (+) and (+++), this can also be written

$$p_\beta(y)p_{\alpha+\beta}(-y) = p_{\alpha+\beta}(zy)p_{\beta}(-z^{-1}y),$$

and since p_β and $p_{\alpha+\beta}$ commute, this gives $z = -1$.

3.2.7. — We have thus proved (ii) and (iii); let us prove (i). One has

$$w_\alpha w_\beta w_\alpha = w_\alpha w_\beta w_\alpha^{-1} w_\alpha^2 = w_{\alpha+\beta}^{-1} t_\alpha,$$

whence

$$\begin{aligned} w_\beta w_\alpha w_\beta w_\alpha w_\beta &= w_\beta w_{\alpha+\beta}^{-1} t_\alpha w_\beta \\ &= w_\beta w_{\alpha+\beta}^{-1} w_\beta^{-1} \cdot s_\beta(t_\alpha) \cdot t_\beta \\ &= w_\alpha \cdot t_\alpha t_\beta \cdot t_\beta = w_\alpha t_\alpha = w_\alpha^{-1}, \end{aligned}$$

which gives

$$(w_\alpha w_\beta)^3 = (w_\beta w_\alpha)^3 = e,$$

which completes the proof.

Remark 3.2.8. Condition (v) of 2.3 is expressed here as follows (writing $v_\alpha = f_\alpha(u_\alpha)$):

$$\begin{aligned} \text{(A)} \quad & h_\alpha v_\beta^{-1} h_\alpha^{-1} = h_\beta v_\alpha h_\beta^{-1}, \\ \text{(B)} \quad & \begin{cases} v_\beta v_\alpha = v_\alpha v_\beta \cdot h_\beta v_\alpha h_\beta^{-1}, \\ v_\beta \cdot h_\beta v_\alpha h_\beta^{-1} = h_\beta v_\alpha h_\beta^{-1} \cdot v_\beta, \\ v_\alpha \cdot h_\beta v_\alpha h_\beta^{-1} = h_\beta v_\alpha h_\beta^{-1} \cdot v_\alpha. \end{cases} \end{aligned}$$

Putting $v_{\alpha+\beta} = \text{int}(h_\beta)v_\alpha$, the last three conditions can also be written

$$\text{(B)} \quad \begin{cases} v_\beta v_\alpha = v_\alpha v_\beta v_{\alpha+\beta}, \\ v_\alpha v_{\alpha+\beta} = v_{\alpha+\beta} v_\alpha, \\ v_\beta v_{\alpha+\beta} = v_{\alpha+\beta} v_\beta. \end{cases}$$

3.3. — Groups of type B_2

Proposition 3.3.1. *Let S be a scheme, G a pinned S -group of type B_2 ; write $\Delta = \{\alpha, \beta\}$, $R_+ = \{\alpha, \beta, \alpha + \beta, 2\alpha + \beta\}$.*

(i) *One has*

$$t_{\alpha\beta} = (w_\alpha w_\beta)^4 = t_\alpha = (w_\beta w_\alpha)^4 = t_{\beta\alpha}.$$

(ii) *If one puts*

$$\text{Ad}(w_\beta)X_\alpha = X_{\alpha+\beta}, \quad \text{Ad}(w_\alpha)X_\beta = X_{2\alpha+\beta},$$

one has:

$$\begin{aligned} \text{Ad}(w_\alpha)X_{\alpha+\beta} &= -X_{\alpha+\beta}, \\ \text{Ad}(w_\alpha)X_{2\alpha+\beta} &= X_\beta, \\ \text{Ad}(w_\beta)X_{\alpha+\beta} &= -X_\alpha, \\ \text{Ad}(w_\beta)X_{2\alpha+\beta} &= X_{2\alpha+\beta}. \end{aligned}$$

(iii) Put

$$\begin{aligned} p_{\alpha+\beta}(x) &= \exp(xX_{\alpha+\beta}) = \text{int}(w_\beta)p_\alpha(x), \\ p_{2\alpha+\beta}(x) &= \exp(xX_{2\alpha+\beta}) = \text{int}(w_\alpha)p_\beta(x). \end{aligned}$$

Then:

$$\begin{aligned} p_\beta(y)p_\alpha(x) &= p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(xy)p_{2\alpha+\beta}(x^2y), \\ p_{\alpha+\beta}(y)p_\alpha(x) &= p_\alpha(x)p_{\alpha+\beta}(y)p_{2\alpha+\beta}(2xy). \end{aligned}$$

3.3.2. — The proof occupies nos. 3.3.2 to 3.3.6. One has $(\beta^*, \alpha) = -1$, $(\alpha^*, \beta) = -2$, whence $\alpha(t_\beta) = -1$, $\beta(t_\alpha) = 1$. Observe in passing that $\beta(t_\alpha) = \alpha(t_\alpha) = 1$, which shows that t_α is a section of $\text{Centr}(G)$. Put

$$\text{Ad}(w_\beta)X_\alpha = X_{\alpha+\beta}, \quad \text{Ad}(w_\alpha)X_\beta = X_{2\alpha+\beta}.$$

One immediately has

$$\begin{aligned} \text{Ad}(w_\beta)X_{\alpha+\beta} &= \alpha(t_\beta)X_\alpha = -X_\alpha, \\ \text{Ad}(w_\alpha)X_{2\alpha+\beta} &= \beta(t_\alpha)X_\beta = X_\beta. \end{aligned}$$

Since $(2\alpha + \beta) + \beta$ and $(2\alpha + \beta) - \beta$ are not roots, one has

$$\text{Ad}(w_\beta)X_{2\alpha+\beta} = X_{2\alpha+\beta}.$$

There exists a scalar $k \in \mathbf{G}_m(S)$ such that

$$\text{Ad}(w_\alpha)X_{\alpha+\beta} = kX_{\alpha+\beta}.$$

On the other hand, by Exp. XXII, 5.5.2, there exist sections $A, B, C \in \mathbf{G}_a(S)$ such that

$$(1) \quad p_\beta(y)p_\alpha(x) = p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(Axy)p_{2\alpha+\beta}(Bx^2y),$$

$$(2) \quad p_{\alpha+\beta}(y)p_\alpha(x) = p_\alpha(x)p_{\alpha+\beta}(y)p_{2\alpha+\beta}(Cxy).$$

Thus, in (ii) and (iii), the task is to prove $A = B = 1$, $C = 2$, $k = -1$.

3.3.3. — Let $\text{int}(w_\alpha)$ act on formula (2). One finds

$$(3) \quad p_{2\alpha+\beta}(y)p_{-\alpha}(-x) = p_{-\alpha}(-x)p_{2\alpha+\beta}(y)p_{\alpha+\beta}(Akxy)p_\beta(Bx^2y).$$

Transforming (2) in the same way, one obtains

$$(4) \quad p_{\alpha+\beta}(ky)p_{-\alpha}(-x) = p_{-\alpha}(-x)p_{\alpha+\beta}(ky)p_\beta(Cxy).$$

Transforming (1) by $\text{int}(w_\beta)$, one has

$$(5) \quad p_{-\beta}(-y)p_{\alpha+\beta}(x) = p_{\alpha+\beta}(x)p_{-\beta}(-y)p_\alpha(-Axy)p_{2\alpha+\beta}(Bx^2y).$$

3.3.4. — Let us write

$$w_\beta p_\alpha(X)w_\beta^{-1} = p_{\alpha+\beta}(x).$$

Since $\alpha + 2\beta$ is not a root, this gives

$$p_\beta(1)p_\alpha(x)p_\beta(-1) = p_{-\beta}(1)p_{\alpha+\beta}(x)p_{-\beta}(-1).$$

Using (1) on the left and (5) on the right, one obtains

$$\begin{aligned} p_\alpha(x)p_\beta(1)p_{\alpha+\beta}(Ax)p_{2\alpha+\beta}(Bx^2)p_\beta(-1) \\ = p_{\alpha+\beta}(x)p_{-\beta}(1)p_\alpha(Ax)p_{2\alpha+\beta}(-Bx^2)p_{-\beta}(-1). \end{aligned}$$

Since p_β commutes with $p_{\alpha+\beta}$ and $p_{2\alpha+\beta}$ on the one hand, and $p_{-\beta}$ commutes with p_α and $p_{2\alpha+\beta}$ on the other hand, this can be written

$$p_\alpha(x)p_{\alpha+\beta}(Ax)p_{2\alpha+\beta}(Bx^2) = p_{\alpha+\beta}(x)p_\alpha(Ax)p_{2\alpha+\beta}(-Bx^2).$$

Transforming the right side by (2), one obtains

$$\begin{aligned} p_\alpha(x)p_{\alpha+\beta}(Ax)p_{2\alpha+\beta}(Bx^2) \\ = p_\alpha(Ax)p_{\alpha+\beta}(x)p_{2\alpha+\beta}((AC - B)x^2), \end{aligned}$$

which gives

$$A = 1, \quad C = 2B.$$

3.3.5. — Let us now write

$$w_\alpha p_\beta(y) w_\alpha^{-1} = p_{2\alpha+\beta}(y).$$

Since $3\alpha + \beta$ is not a root, this gives

$$p_\alpha(1) p_\beta(y) p_\alpha(-1) = p_{-\alpha}(1) p_{2\alpha+\beta}(y) p_{-\alpha}(-1).$$

Using (1) on the left and (3) on the right, one obtains

$$p_\beta(y) p_{\alpha+\beta}(-Ay) p_{2\alpha+\beta}(By) = p_{2\alpha+\beta}(y) p_{\alpha+\beta}(Aky) p_\beta(By).$$

Since $p_{\alpha+\beta}$, $p_{2\alpha+\beta}$ and p_β commute, this immediately gives

$$B = 1, \quad -A = Ak,$$

whence finally

$$A = 1, \quad B = 1, \quad C = 2, \quad k = -1.$$

3.3.6. — We have thus proved (ii) and (iii). Let us prove (i). Taking into account the equality $s_\beta(t_\alpha) = t_\alpha t_\beta^{(\alpha^*, \beta)} = t_\alpha$ (since $(\alpha^*, \beta) = 2$), one successively has:

$$\begin{aligned} w_\alpha w_\beta w_\alpha &= w_\alpha w_\beta w_\alpha^{-1} t_\alpha = w_{2\alpha+\beta} t_\alpha, \\ w_\beta w_\alpha w_\beta w_\alpha w_\beta &= w_\beta w_{2\alpha+\beta} w_\beta^{-1} \cdot s_\beta(t_\alpha) \cdot t_\beta = w_{2\alpha+\beta} t_\alpha t_\beta, \\ w_\alpha w_\beta w_\alpha w_\beta w_\alpha w_\beta w_\alpha &= w_\alpha w_{2\alpha+\beta} w_\alpha^{-1} \cdot s_\alpha(t_\alpha t_\beta) \cdot t_\alpha \\ &= w_\beta \cdot t_\alpha \cdot t_\beta t_\alpha \cdot t_\alpha = w_\beta^{-1} t_\alpha, \end{aligned}$$

whence

$$(w_\beta w_\alpha)^4 = t_\alpha,$$

and

$$(w_\alpha w_\beta)^4 = s_\beta(t_\alpha) = t_\alpha,$$

which completes the proof.

Remark 3.3.7. Condition (v) of 2.3 is expressed here in the following way, putting $v_{\alpha+\beta} = \text{int}(h_\beta)v_\alpha$ and $v_{2\alpha+\beta} = \text{int}(h_\alpha)v_\beta$:

$$\begin{aligned} \text{(A)} \quad & \begin{cases} \text{int}(h_\alpha h_\beta h_\alpha) v_\beta = v_\beta, \\ \text{int}(h_\beta h_\alpha h_\beta) v_\alpha = v_\alpha, \end{cases} \\ \text{(B)} \quad & \begin{cases} v_\beta v_\alpha = v_\alpha v_\beta v_{\alpha+\beta} v_{2\alpha+\beta}, \\ v_{\alpha+\beta} v_\alpha = v_\alpha v_{\alpha+\beta} v_{2\alpha+\beta}^2, \\ v_{\alpha+\beta} v_\beta = v_\beta v_{\alpha+\beta}, \\ v_{2\alpha+\beta} v_\alpha = v_\alpha v_{2\alpha+\beta}, \\ v_{2\alpha+\beta} v_\beta = v_\beta v_{2\alpha+\beta}, \\ v_{2\alpha+\beta} v_{\alpha+\beta} = v_{\alpha+\beta} v_{2\alpha+\beta}. \end{cases} \end{aligned}$$

3.4. — Groups of type G_2

Proposition 3.4.1. *Let S be a scheme, G a pinned S -group of type G_2 ; write $\Delta = \{\alpha, \beta\}$, $R_+ = \{\alpha, \beta, \alpha + \beta, 2\alpha + \beta, 3\alpha + \beta, 3\alpha + 2\beta\}$.*

(i) *One has*

$$t_{\alpha\beta} = (w_\alpha w_\beta)^6 = e = (w_\beta w_\alpha)^6 = t_{\beta\alpha}.$$

(ii) If one puts

$$\begin{aligned} \operatorname{Ad}(w_\beta)X_\alpha &= X_{\alpha+\beta}, & \operatorname{Ad}(w_\alpha)X_{\alpha+\beta} &= X_{2\alpha+\beta}, \\ \operatorname{Ad}(w_\alpha)X_\beta &= -X_{3\alpha+\beta}, & \operatorname{Ad}(w_\beta)X_{3\alpha+\beta} &= X_{3\alpha+2\beta}, \end{aligned}$$

one has:

$$\begin{aligned} \operatorname{Ad}(w_\alpha)X_{2\alpha+\beta} &= -X_{\alpha+\beta}, & \operatorname{Ad}(w_\alpha)X_{3\alpha+\beta} &= X_\beta, \\ \operatorname{Ad}(w_\alpha)X_{3\alpha+2\beta} &= X_{3\alpha+2\beta}, & \operatorname{Ad}(w_\beta)X_{\alpha+\beta} &= -X_\alpha, \\ \operatorname{Ad}(w_\beta)X_{2\alpha+\beta} &= X_{2\alpha+\beta}, & \operatorname{Ad}(w_\beta)X_{3\alpha+2\beta} &= -X_{3\alpha+\beta}. \end{aligned}$$

(iii) If one puts

$$\begin{aligned} p_{\alpha+\beta}(x) &= \exp(xX_{\alpha+\beta}) = \operatorname{int}(w_\beta)p_\alpha(x), \\ p_{2\alpha+\beta}(x) &= \exp(xX_{2\alpha+\beta}) = \operatorname{int}(w_\alpha w_\beta)p_\alpha(x), \\ p_{3\alpha+\beta}(x) &= \exp(xX_{3\alpha+\beta}) = \operatorname{int}(w_\alpha)p_\beta(-x), \\ p_{3\alpha+2\beta}(x) &= \exp(xX_{3\alpha+2\beta}) = \operatorname{int}(w_\beta w_\alpha)p_\beta(-x), \end{aligned}$$

one has:

$$\begin{aligned} p_\beta(y)p_\alpha(x) &= p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(xy)p_{2\alpha+\beta}(x^2y) \\ &\quad \cdot p_{3\alpha+\beta}(x^3y)p_{3\alpha+2\beta}(x^3y^2), \\ p_{\alpha+\beta}(y)p_\alpha(x) &= p_\alpha(x)p_{\alpha+\beta}(y)p_{2\alpha+\beta}(2xy) \\ &\quad \cdot p_{3\alpha+\beta}(3x^2y)p_{3\alpha+2\beta}(3xy^2), \\ p_{2\alpha+\beta}(y)p_\alpha(x) &= p_\alpha(x)p_{2\alpha+\beta}(y)p_{3\alpha+\beta}(3xy), \\ p_{3\alpha+\beta}(y)p_\beta(x) &= p_\beta(x)p_{3\alpha+\beta}(y)p_{3\alpha+2\beta}(-xy), \\ p_{2\alpha+\beta}(y)p_{\alpha+\beta}(x) &= p_{\alpha+\beta}(x)p_{2\alpha+\beta}(y)p_{3\alpha+2\beta}(3xy). \end{aligned}$$

3.4.2. — The proof occupies nos. 3.4.2 to 3.4.9. One has $(\beta^*, \alpha) = -1$, $(\alpha^*, \beta) = -3$, whence $\beta(t_\alpha) = \alpha(t_\beta) = -1$. Define $X_{\alpha+\beta}$, $X_{2\alpha+\beta}$, $X_{3\alpha+\beta}$ and $X_{3\alpha+2\beta}$ as in (ii). One immediately has

$$\begin{aligned} \operatorname{Ad}(w_\beta)X_{\alpha+\beta} &= \alpha(t_\beta)X_\alpha = -X_\alpha, \\ \operatorname{Ad}(w_\alpha)X_{2\alpha+\beta} &= \alpha(t_\alpha)\beta(t_\alpha)X_{\alpha+\beta} = -X_{\alpha+\beta}, \\ \operatorname{Ad}(w_\alpha)X_{3\alpha+\beta} &= -\beta(t_\alpha)X_\beta = X_\beta, \\ \operatorname{Ad}(w_\beta)X_{3\alpha+2\beta} &= \alpha(t_\beta)^3\beta(t_\beta)X_{3\alpha+\beta} = -X_{3\alpha+\beta}. \end{aligned}$$

Finally, since $(3\alpha + 2\beta) \pm \alpha$ and $(2\alpha + \beta) \pm \beta$ are not roots, one has:

$$\operatorname{Ad}(w_\alpha)X_{3\alpha+2\beta} = X_{3\alpha+2\beta}, \quad \operatorname{Ad}(w_\beta)X_{2\alpha+\beta} = X_{2\alpha+\beta},$$

which completes the proof of (ii).

3.4.3. — By Exp. XXII, 5.5.2, there exist scalars⁽⁸⁾

$$A, B, C, D, E, F, G, H, J \in \mathbf{G}_a(S),$$

such that

$$(1) \quad \begin{aligned} p_\beta(y)p_\alpha(x) &= p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(Axy)p_{2\alpha+\beta}(Bx^2y) \\ &\quad p_{3\alpha+\beta}(Cx^3y)p_{3\alpha+2\beta}(Dx^3y^2). \end{aligned}$$

$$(2) \quad \begin{aligned} p_{\alpha+\beta}(y)p_\alpha(x) &= p_\alpha(x)p_{\alpha+\beta}(y)p_{2\alpha+\beta}(Exy) \\ &\quad p_{3\alpha+\beta}(Fx^2y)p_{3\alpha+2\beta}(Gxy^2). \end{aligned}$$

$$(3) \quad p_{2\alpha+\beta}(y)p_\alpha(x) = p_\alpha(x)p_{2\alpha+\beta}(y)p_{3\alpha+\beta}(Hxy).$$

⁽⁸⁾**N.D.E.** Absolute constants A, B, ..., J are introduced here. These constants will be determined in the following pages; their values are $A = B = C = D = 1$, $E = 2$, $J = -1$, $F = G = H = 3$, cf. 3.4.8.

$$(4) \quad p_{3\alpha+\beta}(y)p_{\beta}(x) = p_{\beta}(x)p_{3\alpha+\beta}(y)p_{3\alpha+2\beta}(Jxy).$$

3.4.4. — Let $\text{int}(w_{\beta})$ act on (1), (3) and (4):

$$(5) \quad \begin{aligned} p_{-\beta}(-y)p_{\alpha+\beta}(x) &= p_{\alpha+\beta}(x)p_{-\beta}(-y)p_{\alpha}(-Axy) \\ p_{2\alpha+\beta}(Bx^2y)p_{3\alpha+2\beta}(Cx^3y)p_{3\alpha+\beta}(-Dx^3y^2). \end{aligned}$$

$$(6) \quad p_{2\alpha+\beta}(y)p_{\alpha+\beta}(x) = p_{\alpha+\beta}(x)p_{2\alpha+\beta}(y)p_{3\alpha+2\beta}(Hxy).$$

$$(7) \quad p_{3\alpha+2\beta}(y)p_{-\beta}(-x) = p_{-\beta}(-x)p_{3\alpha+2\beta}(y)p_{3\alpha+\beta}(-Jxy).$$

Similarly letting $\text{int}(w_{\alpha})$ act on (1), one finds

$$(8) \quad \begin{aligned} p_{3\alpha+\beta}(-y)p_{-\alpha}(-x) &= p_{-\alpha}(-x)p_{3\alpha+\beta}(-y)p_{2\alpha+\beta}(Axy) \\ p_{\alpha+\beta}(-Bx^2y)p_{\beta}(Cx^3y)p_{3\alpha+\beta}(Dx^3y^2). \end{aligned}$$

3.4.5. — Let us write

$$w_{\beta}p_{\alpha}(x)w_{\beta}^{-1} = p_{\alpha+\beta}(x),$$

that is, $\alpha + 2\beta$ not being a root:

$$(9) \quad p_{\beta}(1)p_{\alpha}(x)p_{\alpha}(-1) = p_{-\beta}(1)p_{\alpha+\beta}(x)p_{-\beta}(-1).$$

Transforming the left side of (9) by (1), then (4), one obtains:

$$(10) \quad \begin{aligned} &p_{\beta}(1)p_{\alpha}(x)p_{\beta}(-1) \\ &= p_{\alpha}(x)p_{\beta}(1)p_{\alpha+\beta}(Ax)p_{2\alpha+\beta}(Bx^2) \\ &\quad p_{3\alpha+\beta}(Cx^3)p_{3\alpha+2\beta}(Dx^3)p_{\beta}(-1) \\ &= p_{\alpha}(x)p_{\beta}(1)p_{\alpha+\beta}(Ax)p_{2\alpha+\beta}(Bx^2)p_{\beta}(-1) \\ &\quad p_{3\alpha+\beta}(Cx^3)p_{3\alpha+2\beta}((D - CJ)x^3) \\ &= p_{\alpha}(x)p_{\alpha+\beta}(Ax)p_{2\alpha+\beta}(Bx^2)p_{3\alpha+\beta}(Cx^3) \\ &\quad p_{3\alpha+2\beta}((D - CJ)x^3). \end{aligned}$$

Let us transform the right side of (9) by (5), then (7):

$$(11) \quad \begin{aligned} &p_{-\beta}(1)p_{\alpha+\beta}(x)p_{-\beta}(-1) \\ &= p_{\alpha+\beta}(x)p_{-\beta}(1)p_{\alpha}(Ax)p_{2\alpha+\beta}(-Bx^2) \\ &\quad p_{3\alpha+2\beta}(-Cx^3)p_{3\alpha+\beta}(-Dx^3)p_{-\beta}(-1) \\ &= p_{\alpha+\beta}(x)p_{\alpha}(Ax)p_{2\alpha+\beta}(-Bx^2) \\ &\quad p_{3\alpha+2\beta}(-Cx^3)p_{3\alpha+\beta}((CJ - D)x^3). \end{aligned}$$

Now using (2), this right side becomes

$$(12) \quad \begin{aligned} &p_{\alpha}(Ax)p_{\alpha+\beta}(x)p_{2\alpha+\beta}(AEx^2)p_{3\alpha+\beta}(A^2Fx^3)p_{3\alpha+2\beta}(AGx^3) \times \\ &\quad p_{2\alpha+\beta}(-Bx^2)p_{3\alpha+2\beta}(-Cx^3)p_{3\alpha+\beta}((CJ - D)x^3) \\ &= p_{\alpha}(Ax)p_{\alpha+\beta}(x)p_{2\alpha+\beta}((AE - B)x^2) \\ &\quad p_{3\alpha+\beta}((A^2F + CJ - D)x^3)p_{3\alpha+2\beta}((AG - C)x^3). \end{aligned}$$

Thus (9) can be rewritten:

$$\begin{aligned} &p_{\alpha}(x)p_{\alpha+\beta}(Ax)p_{2\alpha+\beta}(Bx^2)p_{3\alpha+\beta}(Cx^3)p_{3\alpha+2\beta}((D - CJ)x^3) \\ &= p_{\alpha}(Ax)p_{\alpha+\beta}(x)p_{2\alpha+\beta}((AE - B)x^2) \\ &\quad p_{3\alpha+\beta}((A^2F + CJ - D)x^3)p_{3\alpha+2\beta}((AG - C)x^3) \end{aligned}$$

which gives

$$A = 1, \quad E = 2B, \quad C + D = F + CJ, \quad F = G.$$

3.4.6. — Let us now write

$$w_\alpha p_\beta(y) w_\alpha^{-1} = p_{3\alpha+\beta}(-y),$$

that is, $4\alpha + \beta$ not being a root:

$$(13) \quad p_\alpha(1) p_\beta(y) p_\alpha(-1) = p_{-\alpha}(1) p_{3\alpha+\beta}(-y) p_{-\alpha}(-1).$$

Let us transform the left side by (1):

$$\begin{aligned} p_\alpha(1) p_\beta(y) p_\alpha(-1) &= p_\beta(y) p_{\alpha+\beta}(-Ay) p_{2\alpha+\beta}(By) \\ &\quad p_{3\alpha+\beta}(-Cy) p_{3\alpha+2\beta}(-Dy^2). \end{aligned}$$

Let us transform the right side of (13) successively by (8), (6) and (4):

$$\begin{aligned} &p_{-\alpha}(1) p_{3\alpha+\beta}(-y) p_{-\alpha}(-1) \\ &= p_{3\alpha+\beta}(-y) p_{2\alpha+\beta}(Ay) p_{\alpha+\beta}(-By) p_\beta(Cy) p_{3\alpha+2\beta}(Dy^2) \\ &= p_{3\alpha+\beta}(-y) p_{\alpha+\beta}(-By) p_{2\alpha+\beta}(Ay) \\ &\quad p_{3\alpha+2\beta}(-ABHy^2) p_\beta(Cy) p_{3\alpha+2\beta}(Dy^2) \\ &= p_\beta(Cy) p_{3\alpha+\beta}(-y) p_{\alpha+\beta}(-By) p_{2\alpha+\beta}(Ay) \\ &\quad p_{3\alpha+2\beta}((D - CJ - ABH)y^2) \\ &= p_\beta(Cy) p_{\alpha+\beta}(-By) p_{2\alpha+\beta}(Ay) p_{3\alpha+\beta}(-y) \\ &\quad p_{3\alpha+2\beta}((D - CJ - ABH)y^2). \end{aligned}$$

Thus (13) can be rewritten:

$$\begin{aligned} &p_\beta(Cy) p_{\alpha+\beta}(-By) p_{2\alpha+\beta}(Ay) p_{3\alpha+\beta}(-y) p_{3\alpha+2\beta}((D - CJ - ABH)y^2) \\ &= p_\beta(y) p_{\alpha+\beta}(-Ay) p_{2\alpha+\beta}(By) p_{3\alpha+\beta}(-Cy) p_{3\alpha+2\beta}(-Dy^2) \end{aligned}$$

whence

$$C = 1, \quad A = B, \quad D - CJ - ABH = -D.$$

Let us take into account the results already obtained:

$$\begin{aligned} A = B = C = 1, \quad E = 2, \quad F = G, \\ D + 1 = F + J, \quad 2D = H + J. \end{aligned}$$

3.4.7. — Let us write

$$w_\beta p_{3\alpha+\beta}(x) w_\beta^{-1} = p_{3\alpha+2\beta}(x),$$

that is

$$p_\beta(1) p_{3\alpha+\beta}(x) p_\beta(-1) = p_{-\beta}(1) p_{3\alpha+2\beta}(x) p_{-\beta}(-1).$$

Transforming the left side by (4), and the right side by (7), one obtains:

$$p_{3\alpha+\beta}(x) p_{3\alpha+2\beta}(-Jx) = p_{3\alpha+2\beta}(x) p_{3\alpha+\beta}(-Jx),$$

that is, $J = -1$.

3.4.8. — Finally, let us write

$$w_\alpha p_{\alpha+\beta}(y) w_\alpha^{-1} = p_{2\alpha+\beta}(y),$$

that is

$$\begin{aligned} &p_\alpha(1) p_{\alpha+\beta}(y) p_\alpha(-1) \\ &= p_{-\alpha}(1) p_\alpha(-1) p_{2\alpha+\beta}(y) p_\alpha(1) p_{-\alpha}(-1). \end{aligned}$$

Transforming the left side by (2), and the right side by (3), one obtains:

$$\begin{aligned} &p_{\alpha+\beta}(y) p_{2\alpha+\beta}(-Ey) p_{3\alpha+\beta}(Fy) p_{3\alpha+2\beta}(-Gy^2) \\ &= p_{-\alpha}(1) p_{2\alpha+\beta}(y) p_{3\alpha+\beta}(Hy) p_{-\alpha}(-1). \end{aligned}$$

It is immediate to see that if one commutes $p_{-\alpha}(-1)$ with $p_{3\alpha+\beta}(Hy)$, then $p_{2\alpha+\beta}(y)$, one introduces no new terms in $p_{3\alpha+\beta}$ on the right side. The latter can thus be written, denoting by empty parentheses the quantities whose exact value does not matter to us:

$$p_{\alpha+\beta}(\)p_{2\alpha+\beta}(\)p_{\beta}(\)p_{3\alpha+\beta}(Hy)p_{3\alpha+2\beta}(\).$$

Comparing with the left side, one immediately has $F = H$, whence, by the earlier results, $2D = D + 1$, that is, $D = 1$, and finally $F = G = H = 2D - J = 3$, which completes the determination of the coefficients $A, \dots, J$ and the proof of (iii).

3.4.9. — Finally, let us prove (i) in the usual manner. One successively has:

$$\begin{aligned} w_{\alpha}w_{\beta}w_{\alpha} &= w_{\alpha}w_{\beta}w_{\alpha}^{-1}t_{\alpha} = w_{3\alpha+\beta}^{-1} \cdot t_{\alpha}, \\ w_{\beta}(w_{\alpha}w_{\beta})^2 &= w_{\beta}w_{3\alpha+\beta}^{-1}w_{\beta}^{-1} \cdot s_{\beta}(t_{\alpha}) \cdot t_{\beta} \\ &= w_{3\alpha+2\beta}^{-1} \cdot t_{\alpha}t_{\beta} \cdot t_{\beta} = w_{3\alpha+2\beta}^{-1} \cdot t_{\alpha}, \\ w_{\alpha}(w_{\beta}w_{\alpha})^3 &= w_{\alpha}w_{3\alpha+2\beta}^{-1}w_{\alpha}^{-1} = w_{3\alpha+2\beta}^{-1}, \\ w_{\beta}(w_{\alpha}w_{\beta})^4 &= w_{\beta}w_{3\alpha+2\beta}^{-1}w_{\beta}^{-1} \cdot t_{\beta} = w_{3\alpha+\beta} \cdot t_{\beta}, \\ w_{\alpha}(w_{\beta}w_{\alpha})^5 &= w_{\alpha}w_{3\alpha+\beta}w_{\alpha}^{-1} \cdot s_{\alpha}(t_{\beta}) \cdot t_{\beta} \\ &= w_{\beta} \cdot t_{\beta}t_{\alpha} \cdot t_{\alpha} = w_{\beta}^{-1}. \end{aligned}$$

Whence

$$(w_{\alpha}w_{\beta})^6 = (w_{\beta}w_{\alpha})^6 = e.$$

Remark 3.4.10. Condition (v) of 2.4 consists of

$$(A) \quad \begin{cases} \text{int}(h_{\alpha}h_{\beta}h_{\alpha}h_{\beta}h_{\alpha})v_{\beta} = v_{\beta}, \\ \text{int}(h_{\beta}h_{\alpha}h_{\beta}h_{\alpha}h_{\beta})v_{\alpha} = v_{\alpha}, \end{cases}$$

and, putting

$$\begin{aligned} v_{\alpha+\beta} &= \text{int}(h_{\beta})v_{\alpha}, & v_{2\alpha+\beta} &= \text{int}(h_{\alpha}h_{\beta})v_{\alpha}, \\ v_{3\alpha+\beta} &= \text{int}(h_{\alpha})v_{\beta}^{-1}; & v_{3\alpha+2\beta} &= \text{int}(h_{\beta}h_{\alpha})v_{\beta}^{-1}, \end{aligned}$$

of the commutation relations:

$$(B) \quad \begin{cases} v_{\beta}v_{\alpha} = v_{\alpha}v_{\beta}v_{\alpha+\beta}v_{2\alpha+\beta}v_{3\alpha+\beta}v_{3\alpha+2\beta}, \\ v_{\alpha+\beta}v_{\alpha} = v_{\alpha}v_{\alpha+\beta}v_{2\alpha+\beta}^2v_{3\alpha+\beta}^3v_{3\alpha+2\beta}^3, \\ v_{2\alpha+\beta}v_{\alpha} = v_{\alpha}v_{2\alpha+\beta}v_{3\alpha+\beta}^3, \\ v_{3\alpha+\beta}v_{\alpha} = v_{\alpha}v_{3\alpha+\beta}, \\ v_{3\alpha+2\beta}v_{\alpha} = v_{\alpha}v_{3\alpha+2\beta}, \end{cases}$$

$$(B) \quad \begin{cases} v_{\alpha+\beta}v_{\beta} = v_{\beta}v_{\alpha+\beta}, \\ v_{2\alpha+\beta}v_{\beta} = v_{\beta}v_{2\alpha+\beta}, \\ v_{3\alpha+\beta}v_{\beta} = v_{\beta}v_{3\alpha+\beta}v_{3\alpha+2\beta}^{-1}, \\ v_{3\alpha+2\beta}v_{\beta} = v_{\beta}v_{3\alpha+2\beta}, \\ v_{2\alpha+\beta}v_{\alpha+\beta} = v_{\alpha+\beta}v_{2\alpha+\beta}v_{3\alpha+2\beta}^3, \end{cases}$$

$$(B) \quad \begin{cases} v_{3\alpha+\beta}v_{\alpha+\beta} = v_{\alpha+\beta}v_{3\alpha+\beta}, \\ v_{3\alpha+2\beta}v_{\alpha+\beta} = v_{\alpha+\beta}v_{3\alpha+2\beta}, \\ v_{3\alpha+\beta}v_{2\alpha+\beta} = v_{2\alpha+\beta}v_{3\alpha+\beta}, \\ v_{3\alpha+2\beta}v_{2\alpha+\beta} = v_{2\alpha+\beta}v_{3\alpha+2\beta}, \\ v_{3\alpha+2\beta}v_{3\alpha+\beta} = v_{3\alpha+\beta}v_{3\alpha+2\beta}. \end{cases}$$

3.5. — Explicit form of the theorem on generators and relations

Theorem 3.5.1. *Let S be a scheme, G a pinned S -group, T its maximal torus, Δ its system of simple roots, and $u_\alpha \in U_\alpha(S)^\times$ and $w_\alpha \in \underline{\text{Norm}}_G(T)(S)$ the elements defined by the pinning ($\alpha \in \Delta$). Let*

$$f_T : T \longrightarrow H, \quad f_\alpha : U_\alpha \longrightarrow H, \quad \alpha \in \Delta$$

be group morphisms, H being an S -group sheaf for fppf; let $h_\alpha \in H(S)$, ($\alpha \in \Delta$) be sections of H ; put $v_\alpha = f_\alpha(u_\alpha)$, $\alpha \in \Delta$. For there to exist a group morphism

$$f : G \longrightarrow H$$

extending f_T , f_α ($\alpha \in \Delta$) and satisfying $f(w_\alpha) = h_\alpha$ ($\alpha \in \Delta$) (and then necessarily unique), it is necessary and sufficient that the following conditions hold:

(i) *For every $S' \longrightarrow S$, every $\alpha \in \Delta$, every $t \in T(S')$ and every $x \in U_\alpha(S')$, one has*

$$(1) \quad \text{int}(f_T(t))f_\alpha(x) = f_\alpha(\text{int}(t)x) = f_\alpha(x^{\alpha(t)}).$$

(ii) *For every $\alpha \in \Delta$, every $S' \longrightarrow S$ and every $t \in T(S')$, one has*

$$(2) \quad \text{int}(h_\alpha)f_T(t) = f_T(s_\alpha(t)) = f_T(t \cdot \alpha^* \alpha(t)^{-1}).$$

(iii) *For every $\alpha \in \Delta$, one has⁽⁹⁾*

$$(3_1) \quad h_\alpha^2 = f_T(\alpha^*(-1)),$$

$$(4) \quad (h_\alpha v_\alpha)^3 = e.$$

(iv) *For all $\alpha, \beta \in \Delta$, $\alpha \neq \beta$, such that $(\alpha^*, \beta) = 0$ (resp. $(\alpha^*, \beta) = -1$, resp. $(\alpha^*, \beta) = -2$, resp. $(\alpha^*, \beta) = -3$), one has:*

(a) *the relation*

$$(3_2) \quad \begin{cases} (h_\alpha h_\beta)^2 = f_T(\alpha^*(-1)\beta^*(-1)) & \text{if } (\alpha^*, \beta) = 0; \\ (h_\alpha h_\beta)^3 = e, & \text{if } (\alpha^*, \beta) = -1; \\ (h_\alpha h_\beta)^4 = f_T(\alpha^*(-1)), & \text{if } (\alpha^*, \beta) = -2; \\ (h_\alpha h_\beta)^6 = e, & \text{if } (\alpha^*, \beta) = -3. \end{cases}$$

(b) *The relations (A) and (B) of 3.1.3 (resp. 3.2.8, resp. 3.3.7, resp. 3.4.10).*

This follows immediately from 2.3, 2.6 and the calculations made in each particular case.

Remark 3.5.2. One may present the preceding results in a slightly different manner: one gives morphisms, for $\alpha \in \Delta$,

$$a_\alpha : T \cdot U_\alpha \longrightarrow H, \quad b_\alpha : \underline{\text{Norm}}_{Z_\alpha}(T) \longrightarrow H,$$

and one puts

$$h_\alpha = b_\alpha(w_\alpha), \quad v_\alpha = a_\alpha(u_\alpha);$$

then the conditions to be checked are the following:

- (1) all the a_α ($\alpha \in \Delta$) and all the b_α ($\alpha \in \Delta$) have the same restriction to T ;
- (2) conditions (4) and (iv) of 3.5.1 above hold.

3.5.3. — Various applications of this theorem will be given in the following exposé. Let us mention one here: Theorem 3.5.1 gives a description by generators and relations of G in the category of S -sheaves for fppf; in other words, consider for each $S' \longrightarrow S$ the group $H(S')$ generated by $T(S')$, $U_\alpha(S')$, $\alpha \in R$, and w_α , $\alpha \in R$, subject to relations analogous to (1), (2), (3₁), (4), (3₂), (A), (B); then G is precisely the sheaf associated with the presheaf $S' \longmapsto H(S')$.

In particular, if S' is the spectrum of an algebraically closed field k , one has $G(S') = H(S')$ (an immediate consequence of the Nullstellensatz in the form: “an fppf covering sieve of an

⁽⁹⁾**N.D.E.** Relations (3₁) and (3₂) form the description of the normalizer of the torus, (3₁) being, like (4), in a rank 1 group, whereas (3₂) is in a rank 2 group.

algebraically closed field is trivial”), so that 3.5.1 immediately gives an explicit description by generators and relations of the “abstract” group $G(k)$.⁽¹⁰⁾

4. Uniqueness of pinned groups: fundamental theorem

Theorem 4.1. *Let S be a nonempty scheme. The functor $\mathcal{R}$ of 1.6 is fully faithful: let G and G' be two pinned S -groups, p ⁽¹¹⁾ an integer > 0 such that $x \mapsto x^p$ is an endomorphism of $\mathbf{G}_{a,S}$, and $h : \mathcal{R}(G') \rightarrow \mathcal{R}(G)$ a p -morphism of pinned root data. There exists a unique morphism of pinned groups*

$$f : G \rightarrow G'$$

such that $\mathcal{R}(f) = h$.

Uniqueness is proved in 1.9. It suffices to prove existence. By hypothesis, one has a bijection $d : R \xrightarrow{\sim} R'$ and a map $\mathbf{q} : R \rightarrow \{p^n \mid n \in \mathbf{N}\}$ such that

$$h(d(\alpha)) = \mathbf{q}(\alpha)\alpha \quad \text{and} \quad {}^th(\alpha^*) = \mathbf{q}(\alpha)d(\alpha)^*$$

for every $\alpha \in R$. In particular, the semisimple ranks of G and G' coincide.

4.1.1. — Suppose $\text{rgss}(G) = \text{rgss}(G') = 0$. Then G and G' are tori: one has $G = T = D_S(M)$, $G' = T' = D_S(M')$ and h is simply a morphism of ordinary groups $h : M' \rightarrow M$. One then takes $f = D_S(h)$.

4.1.2. — Suppose $\text{rgss}(G) = \text{rgss}(G') = 1$. Then consider

$$f_T = D_S(h) : T \rightarrow T'.$$

By hypothesis one has a commutative diagram, where $\alpha' = d(\alpha)$:

$$\begin{array}{ccccc} \mathbf{G}_{m,S} & \xrightarrow{\alpha^*} & T & \xrightarrow{\alpha} & \mathbf{G}_{m,S} \\ \mathbf{q}(\alpha) \downarrow & & \downarrow f_T & & \downarrow \mathbf{q}(\alpha) \\ \mathbf{G}_{m,S} & \xrightarrow{\alpha'^*} & T' & \xrightarrow{\alpha'} & \mathbf{G}_{m,S}. \end{array}$$

One then applies Exp. XX, 4.1.

4.1.3. — Suppose $\text{rgss}(G) = \text{rgss}(G') = 2$. Then, by Exp. XXI, 7.5.3, one knows all the possibilities for $h : \mathcal{R}(G') \rightarrow \mathcal{R}(G)$. Let us study them successively, checking each time the conditions of 2.5.

Write $\Delta = \{\alpha, \beta\}$, $\Delta' = \{\alpha', \beta'\}$ so that $d(\alpha) = \alpha'$, $d(\beta) = \beta'$.

4.1.4. — G and G' of type $A_1 \times A_1$. One then has

$$h(\alpha') = q\alpha, \quad h(\beta') = q_1\beta.$$

Let us show that the conditions of 2.5 hold: (ii) and (iii) follow from 3.1.2 (ii) and (iii); let us prove (i). One has

$$\begin{aligned} D_S(h)t_{\alpha\beta} &= D_S(h)(t_\alpha t_\beta) = {}^th(\alpha^*)(-1) \cdot {}^th(\beta^*)(-1) \\ &= \alpha'^*((-1)^q) \cdot \beta'^*((-1)^{q_1}). \end{aligned}$$

⁽¹²⁾Now the hypothesis that $x \mapsto x^p$ is an endomorphism of $\mathbf{G}_{a,S}$ (and $S \neq \emptyset$) implies that $p = 1$ or that S has characteristic p ; in every case one has $(-1)^q = -1$ (if q is even, $p = 2$

⁽¹⁰⁾**N.D.E.** For an arbitrary field k and simply connected G , R. Steinberg gave a presentation of the group $G(k)$ in [St62], Th. 3.2; see also [St67], §6, Th. 8.

⁽¹¹⁾**N.D.E.** To avoid a problem of notation later, we have replaced q here by p , so that in what follows, q (resp. q_1) is an arbitrary power of p .

⁽¹²⁾**N.D.E.** We have expanded what follows.

and $1 = -1$). Consequently,

$$D_S(h)t_{\alpha\beta} = \alpha'^*(-1) \cdot \beta'^*(-1) = t'_{\alpha'\beta'}.$$

This shows that condition 2.5 (i) holds.

4.1.5. — G and G' of type A_2 . One then has

$$h(\alpha') = q\alpha, \quad h(\beta') = q\beta.$$

Put $X_{\alpha+\beta} = \text{Ad}(w_\beta)X_\alpha$ and $X'_{\alpha'+\beta'} = \text{Ad}(w_{\beta'})X_{\alpha'}$. Let us check the conditions of 2.5. For (i), one argues as above, using 3.2.1 (i); for (ii), this is immediate by 3.2.1 (ii); it remains to check (iii). One must check that

$$p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(z) \mapsto p'_{\alpha'}(x^q)p'_{\beta'}(y^q)p'_{\alpha'+\beta'}(z^q)$$

is a group morphism. The only nontrivial commutation relation is that of 3.2.1 (iii), which can be written

$$\begin{aligned} p_\beta(y)p_\alpha(x) &= p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(xy), \\ p'_{\beta'}(y^q)p'_{\alpha'}(x^q) &= p'_{\alpha'}(x^q)p'_{\beta'}(y^q)p'_{\alpha'+\beta'}(x^qy^q). \end{aligned}$$

4.1.6. — One argues in the same way for G and G' of type B_2 (resp. G_2), when the root exponents are equal, using 3.3.1 (resp. 3.4.1); to finish the case of rank 2 groups, it therefore remains to treat the two exceptional cases of Exp. XXI, 7.5.3.

4.1.7. — G and G' are of type B_2 , S has characteristic 2, and one has $\mathbf{q}(\alpha) = 2q$, $\mathbf{q}(\beta) = q$. The positive roots are $\{\alpha, \beta, \alpha + \beta, 2\alpha + \beta\}$ and $\{\alpha', \beta', \alpha' + \beta', \alpha' + 2\beta'\}$ (observe that the “short” simple roots are α and β'). One has

$$\begin{aligned} h(\alpha') &= 2q\alpha, & h(\beta') &= q\beta, \\ h(\alpha' + \beta') &= q(2\alpha + \beta), & h(\alpha' + 2\beta') &= 2q(\alpha + \beta), \end{aligned}$$

which gives

$$\begin{aligned} d(\alpha + \beta) &= \alpha' + 2\beta', & \mathbf{q}(\alpha + \beta) &= 2q, \\ d(2\alpha + \beta) &= \alpha' + \beta', & \mathbf{q}(2\alpha + \beta) &= q. \end{aligned}$$

Put

$$\begin{aligned} X_{\alpha+\beta} &= \text{Ad}(w_\beta)X_\alpha, & X_{2\alpha+\beta} &= \text{Ad}(w_\alpha)X_\beta, \\ X'_{\alpha'+\beta'} &= \text{Ad}(w'_{\alpha'})X'_{\beta'}, & X'_{\alpha'+2\beta'} &= \text{Ad}(w'_{\beta'})X'_{\alpha'}. \end{aligned}$$

Let us now check the conditions of 2.5.

(i) Since S has characteristic 2, one has $-1 = 1$ on S , hence $t_{\alpha\beta} = t_\alpha = \alpha^*(-1) = e = \beta'^*(-1) = t'_{\beta'} = t'_{\alpha'\beta'}$ (cf. 3.3.1 (i)).

(ii) By construction one has

$$\begin{aligned} \text{Ad}(w_\alpha)X_\beta &= X_{2\alpha+\beta}, & \text{Ad}(w'_{\alpha'})X'_{\beta'} &= X'_{\alpha'+\beta'} = X'_{d(2\alpha+\beta)}; \\ \text{Ad}(w_\beta)X_\alpha &= X_{\alpha+\beta}, & \text{Ad}(w'_{\beta'})X'_{\alpha'} &= X'_{\alpha'+2\beta'} = X'_{d(\alpha+\beta)}. \end{aligned}$$

By 3.3.1 (ii) and the fact that $-1 = 1$ on S , one has on either side

$$\begin{aligned} \text{Ad}(w_\alpha)X_{\alpha+\beta} &= X_{\alpha+\beta}, \\ \text{Ad}(w'_{\alpha'})X'_{d(\alpha+\beta)} &= \text{Ad}(w'_{\alpha'})X'_{\alpha'+2\beta'} = X'_{\alpha'+2\beta'} = X'_{d(\alpha+\beta)}; \\ \text{Ad}(w_\alpha)X_{2\alpha+\beta} &= X_\beta, \\ \text{Ad}(w'_{\alpha'})X'_{d(2\alpha+\beta)} &= \text{Ad}(w'_{\alpha'})X'_{\alpha'+\beta'} = X'_{\beta'} = X'_{d(\beta)}; \\ \text{Ad}(w_\beta)X_{\alpha+\beta} &= X_\alpha, \\ \text{Ad}(w'_{\beta'})X'_{d(\alpha+\beta)} &= \text{Ad}(w'_{\beta'})X'_{\alpha'+2\beta'} = X'_{\alpha'} = X'_{d(\alpha)}; \\ \text{Ad}(w_\beta)X_{2\alpha+\beta} &= X_{2\alpha+\beta}, \\ \text{Ad}(w'_{\beta'})X'_{d(2\alpha+\beta)} &= \text{Ad}(w'_{\beta'})X'_{\alpha'+\beta'} = X'_{\alpha'+\beta'} = X'_{d(2\alpha+\beta)}. \end{aligned}$$

(iii) By 3.3.1 (iii), one sees that the only nontrivial commutation relation in U (resp. U') is

$$p_\beta(y)p_\alpha(x) = p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(xy)p_{2\alpha+\beta}(x^2y),$$

resp.

$$p'_{\alpha'}(y')p'_{\beta'}(x') = p'_{\beta'}(x')p'_{\alpha'}(y')p'_{\alpha'+\beta'}(x'y')p'_{\alpha'+2\beta'}(x'^2y').$$

We must check that the morphism

$$\begin{aligned} p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(z)p_{2\alpha+\beta}(t) &\longmapsto p'_{\alpha'}(x^{2q})p'_{\beta'}(y^q) \\ &\quad p'_{\alpha'+2\beta'}(z^{2q})p'_{\alpha'+\beta'}(t^q) \end{aligned}$$

is a group morphism; one immediately sees that this amounts to seeing that

$$\begin{aligned} p'_{\beta'}(y^q)p'_{\alpha'}(x^{2q}) &= p'_{\alpha'}(x^{2q})p'_{\beta'}(y^q) \\ p'_{\alpha'+2\beta'}((xy)^{2q})p'_{\alpha'+\beta'}((x^2y)^q), \end{aligned}$$

which is precisely the second relation above (putting $y' = x^{2q}$, $x' = y^q$).

4.1.8. — G and G' are of type G_2 , S has characteristic 3, and one has $\mathbf{q}(\alpha) = 3q$ and $\mathbf{q}(\beta) = q$. The positive roots are $\{\alpha, \beta, \alpha + \beta, 2\alpha + \beta, 3\alpha + \beta, 3\alpha + 2\beta\}$ on the one hand, $\{\alpha', \beta', \alpha' + \beta', \alpha' + 2\beta', \alpha' + 3\beta', 2\alpha' + 3\beta'\}$ on the other hand (as in the preceding case, the short simple roots are α and β'). One has

$$\begin{aligned} h(\alpha') &= 3q\alpha, & h(\beta') &= q\beta, & h(\alpha' + \beta') &= q(3\alpha + \beta), \\ h(\alpha' + 2\beta') &= q(3\alpha + 2\beta), & h(\alpha' + 3\beta') &= 3q(\alpha + \beta), \\ h(2\alpha' + 3\beta') &= 3q(2\alpha + \beta), \end{aligned}$$

which gives

$$\begin{aligned} d(\alpha + \beta) &= \alpha' + 3\beta', & \mathbf{q}(\alpha + \beta) &= 3q, \\ d(2\alpha + \beta) &= 2\alpha' + 3\beta', & \mathbf{q}(2\alpha + \beta) &= 3q, \\ d(3\alpha + \beta) &= \alpha' + \beta', & \mathbf{q}(3\alpha + \beta) &= q, \\ d(3\alpha + 2\beta) &= \alpha' + 2\beta', & \mathbf{q}(3\alpha + 2\beta) &= q. \end{aligned}$$

Put

$$\begin{aligned} X_{\alpha+\beta} &= \text{Ad}(w_\beta)X_\alpha, & X_{2\alpha+\beta} &= \text{Ad}(w_\alpha)X_{\alpha+\beta}, \\ X_{3\alpha+\beta} &= -\text{Ad}(w_\alpha)X_\beta, & X_{3\alpha+2\beta} &= \text{Ad}(w_\beta)X_{3\alpha+\beta}; \\ X'_{\alpha'+\beta'} &= -\text{Ad}(w'_{\alpha'})X'_{\beta'}, & X'_{\alpha'+2\beta'} &= \text{Ad}(w'_{\beta'})X'_{\alpha'+\beta'}, \\ X'_{\alpha'+3\beta'} &= \text{Ad}(w'_{\beta'})X'_{\alpha'}, & X'_{2\alpha'+3\beta'} &= \text{Ad}(w'_{\alpha'})X'_{\alpha'+3\beta'}. \end{aligned}$$

Let us now check the conditions of 2.5.

(i) One has $t_{\alpha\beta} = e$ and $t'_{\alpha'\beta'} = e$ by 3.4.1 (i).

(ii) By construction one has

$$\begin{aligned} \text{Ad}(w_\beta)X_\alpha &= X_{\alpha+\beta}, & \text{Ad}(w'_{\beta'})X'_{\alpha'} &= X'_{\alpha'+3\beta'} = X'_{d(\alpha+\beta)}; \\ \text{Ad}(w_\alpha)X_\beta &= -X_{3\alpha+\beta}, & \text{Ad}(w'_{\alpha'})X'_{\beta'} &= -X'_{\alpha'+\beta'} = -X'_{d(3\alpha+\beta)}; \\ \text{Ad}(w_\beta)X_{3\alpha+\beta} &= X_{3\alpha+2\beta}, \\ \text{Ad}(w'_{\beta'})X'_{d(3\alpha+\beta)} &= \text{Ad}(w'_{\beta'})X'_{\alpha'+\beta'} = X'_{\alpha'+2\beta'} = X'_{d(3\alpha+2\beta)}; \\ \text{Ad}(w_\alpha)X_{\alpha+\beta} &= X_{2\alpha+\beta}, \\ \text{Ad}(w'_{\alpha'})X'_{d(\alpha+\beta)} &= \text{Ad}(w'_{\alpha'})X'_{\alpha'+3\beta'} = X'_{2\alpha'+3\beta'} = X'_{d(2\alpha+\beta)}. \end{aligned}$$

By 3.4.1 (ii), one has on either side:

$$\begin{aligned}
& \text{Ad}(w_\alpha)X_{2\alpha+\beta} = -X_{\alpha+\beta}, \\
& \text{Ad}(w'_{\alpha'})X'_{d(2\alpha+\beta)} = \text{Ad}(w'_{\alpha'})X'_{2\alpha'+3\beta'} = -X'_{\alpha'+3\beta'} = -X'_{d(\alpha+\beta)}; \\
& \quad \dots \quad \dots \\
& \text{Ad}(w_\beta)X_{3\alpha+2\beta} = -X_{3\alpha+\beta}, \\
& \text{Ad}(w'_{\beta'})X'_{d(3\alpha+2\beta)} = \text{Ad}(w'_{\beta'})X'_{\alpha'+2\beta'} = -X'_{\alpha'+\beta'} = -X'_{d(3\alpha+\beta)}.
\end{aligned}$$

(The dots replace four checks of the same kind.)

(iii) The only nontrivial commutation relations in U and U' are, by 3.4.1 (iii) (and taking account of $3 = 0$ on S):

$$\begin{aligned}
p_\beta(y)p_\alpha(x) &= p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(xy)p_{2\alpha+\beta}(x^2y) \\
&\quad p_{3\alpha+\beta}(x^3y)p_{3\alpha+2\beta}(x^3y^2), \\
p_{\alpha+\beta}(y)p_\alpha(x) &= p_\alpha(x)p_{\alpha+\beta}(y)p_{2\alpha+\beta}(-xy), \\
p_{3\alpha+\beta}(y)p_\beta(x) &= p_\beta(x)p_{3\alpha+\beta}(y)p_{3\alpha+2\beta}(-xy); \\
p'_{\alpha'}(y')p'_{\beta'}(x') &= p'_{\beta'}(x')p'_{\alpha'}(y')p'_{\alpha'+\beta'}(-x'y') \\
&\quad p'_{\alpha'+2\beta'}(-x'^2y')p'_{\alpha'+3\beta'}(-x'^3y')p'_{2\alpha'+3\beta'}(-x'^3y'^2), \\
p'_{\alpha'+\beta'}(y')p'_{\beta'}(x') &= p'_{\beta'}(x')p'_{\alpha'+\beta'}(y')p'_{\alpha'+2\beta'}(x'y'), \\
p'_{\alpha'+3\beta'}(y')p'_{\alpha'}(x') &= p'_{\alpha'}(x')p'_{\alpha'+3\beta'}(y')p'_{2\alpha'+3\beta'}(x'y').
\end{aligned}$$

We must check that the morphism φ from U to U' defined by

$$\begin{aligned}
& \varphi(p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(t)p_{2\alpha+\beta}(u)p_{3\alpha+\beta}(v)p_{3\alpha+2\beta}(w)) \\
&= p'_{\alpha'}(x^{3q})p'_{\beta'}(y^q)p'_{\alpha'+3\beta'}(t^{3q})p'_{2\alpha'+3\beta'}(u^{3q}) \\
&\quad p'_{\alpha'+\beta'}(v^q)p'_{\alpha'+2\beta'}(w^q)
\end{aligned}$$

is a group morphism. Now one immediately checks that the last three commutation relations can also be written

$$\begin{aligned}
p'_{\beta'}(y^q)p'_{\alpha'}(x^{3q}) &= p'_{\alpha'}(x^{3q})p'_{\beta'}(y^q)p'_{\alpha'+3\beta'}((xy)^{3q}) \\
&\quad p'_{2\alpha'+3\beta'}((x^2y)^{3q})p'_{\alpha'+\beta'}((x^3y)^q)p'_{\alpha'+2\beta'}((x^3y^2)^q), \\
p'_{\alpha'+3\beta'}(y^{3q})p'_{\alpha'}(x^{3q}) &= p'_{\alpha'}(x^{3q})p'_{\alpha'+3\beta'}(y^{3q})p'_{2\alpha'+3\beta'}(-(xy)^{3q}), \\
p'_{\alpha'+\beta'}(y^q)p'_{\beta'}(x^q) &= p'_{\beta'}(x^q)p'_{\alpha'+\beta'}(y^q)p'_{\alpha'+2\beta'}(-(xy)^q);
\end{aligned}$$

which shows that φ is indeed a group morphism and completes the proof of 4.1.7.

4.1.9. — Case where G and G' have semisimple rank > 2 . For each root $\alpha \in \Delta$, write $\alpha' = d(\alpha) \in \Delta' = d(\Delta)$. For each $(\alpha, \beta) \in \Delta \times \Delta$, consider the pinned groups of semisimple rank ≤ 2 , $Z_{\alpha\beta}$ and $Z'_{\alpha'\beta'}$. The group morphism $M' \rightarrow M$ underlying h defines a p -morphism of root data

$$h_{\alpha\beta} : \mathcal{R}(Z_{\alpha\beta}) \rightarrow \mathcal{R}(Z'_{\alpha'\beta'}).$$

By the preceding results, there thus exists a morphism of pinned groups

$$f_{\alpha\beta} : Z_{\alpha\beta} \rightarrow Z'_{\alpha'\beta'}$$

such that $\mathcal{R}(f_{\alpha\beta}) = h_{\alpha\beta}$. Let us prove that the $f_{\alpha\beta}$ satisfy the gluing condition of 2.5; indeed, $f_{\alpha\beta}|_{Z_\alpha}$ and $f_{\alpha\alpha}$ are two morphisms of pinned groups

$$Z_\alpha \rightarrow Z'_{\alpha'}$$

corresponding to the same morphism of pinned root data, and hence coincide by the uniqueness result already proved. By 2.5, there thus exists a group morphism

$$f : G \rightarrow G'$$

extending the $f_{\alpha\beta}$. This is obviously a morphism of pinned groups such that $\mathcal{R}(f) = h$, which completes the proof of Theorem 4.1.

5. Corollaries of the fundamental theorem

The most important is:

Corollary 5.1. *Let S be a nonempty scheme, G and G' two pinned S -groups, and h an isomorphism of pinned root data*

$$h : \mathcal{R}(G') \xrightarrow{\sim} \mathcal{R}(G).$$

There exists a unique isomorphism of pinned S -groups

$$f : G \xrightarrow{\sim} G'$$

such that $\mathcal{R}(f) = h$.

Observe that 5.1 also follows from 3.5.1 (the relations of 3.5.1 can be written using only the datum of $\mathcal{R}(G)$); observe also that 5.1 follows from the most elementary part of the proof of 4.1 (one need not consider the “exceptional isogenies” of 4.1.7 and 4.1.8).

Corollary 5.2. (“Uniqueness theorem”). *Let S be a scheme, G and G' two splittable S -groups (Exp. XXII, 1.13). If G and G' have the same type (Exp. XXII, 2.6), they are isomorphic.*

Corollary 5.3. *Let S be a scheme, G and G' two splittable S -groups. The following conditions are equivalent:*

- (i) G and G' are isomorphic.
- (ii) G and G' are locally isomorphic for the fpqc topology.
- (iii) There exists an $s \in S$ such that the s -groups G_s and G'_s have the same type.

PROOF. Indeed, one obviously has (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii). On the other hand, (iii) implies that $\mathcal{R}(G) = \mathcal{R}(G_s) = \mathcal{R}(G'_s) = \mathcal{R}(G')$, hence that G and G' satisfy the condition of 5.2. $\square$

Corollary 5.4. (“Uniqueness of Chevalley schemes”). *Let G and G' be two reductive groups over $\mathbf{Z}$ possessing split maximal tori.* The following conditions are equivalent:*

- (i) G and G' are isomorphic.
- (ii) There exists $s \in \text{Spec}(\mathbf{Z})$ such that G_s and G'_s have the same type.
- (iii) $G_{\mathbf{C}}$ and $G'_{\mathbf{C}}$ have the same type.

Indeed, G and G' are splittable by Exp. XXII, 2.2.

Corollary 5.5. (“Existence of outer automorphisms”). *Let S be a scheme, G a pinned S -group, and h an automorphism of the pinned root datum $\mathcal{R}(G)$. There exists a unique automorphism u of G respecting its pinning and such that $\mathcal{R}(u) = h$.*

Let us make the preceding corollary explicit:

Corollary 5.5.bis. *Let S be a scheme, G a split reductive S -group, and R_+ a system of positive roots of G . For each simple root α , choose an isomorphism of vector groups $p_\alpha : \mathbf{G}_{a,S} \xrightarrow{\sim} U_\alpha$. Let h be an automorphism of M permuting the positive roots and the corresponding coroots: if $\alpha \in R_+$, $h(\alpha) \in R_+$ and $h^\vee(\alpha^*) = h(\alpha)^*$. There exists a unique automorphism u of G inducing $D_S(h)$ on T and such that $u \circ p_\alpha = p_{h(\alpha)}$ for every simple root α .*

Corollary 5.6. *Let G and G' be two reductive S -groups. The following conditions are equivalent:*

*In fact, one can prove that every $\mathbf{Z}$ -torus is split.¹³

⁽¹³⁾**N.D.E.** This follows from the fact that every $\mathbf{Z}$ -torus is isotrivial (Exp. X, 5.16) and that every étale covering of $\text{Spec}(\mathbf{Z})$ is trivial.

- (i) G and G' are locally isomorphic for the fpqc topology.
- (i bis) G and G' are locally isomorphic for the étale topology.
- (ii) For every $s \in S$, $G_{\bar{s}}$ and $G'_{\bar{s}}$ are isomorphic.
- (iii) The functions $s \mapsto \text{type of } G_{\bar{s}}$ and $s \mapsto \text{type of } G'_{\bar{s}}$ are equal.

PROOF. Indeed, (i bis) $\Rightarrow$ (i) trivially, (i) $\Rightarrow$ (ii) by the principle of finite extension (EGA IV₃, 9.1.4), (ii) $\Rightarrow$ (iii) trivially; it remains to prove (iii) $\Rightarrow$ (i bis). Now one may suppose G and G' splittable (Exp. XXII, 2.3), in which case the assertion follows from 5.3. $\square$

Corollary 5.7. *Let S be a scheme, G a reductive S -group, and G' an affine, smooth S -group with connected fibers. Let $s \in S$ be such that $G_{\bar{s}}$ and $G'_{\bar{s}}$ are isomorphic; there then exists an $S' \rightarrow S$ which is étale and covers s such that $G_{S'}$ and $G'_{S'}$ are isomorphic.*

Indeed, by Exp. XIX 2.5 and Exp. XXII 2.1, one may suppose G and G' splittable reductive groups, and one is reduced to 5.3.

When S is the spectrum of a field, one deduces from 5.6 and 5.7:

Corollary 5.8. *Let k be a field and G and G' two reductive k -groups. The following conditions are equivalent:⁽¹⁴⁾*

- (i) G and G' have the same type.
- (ii) $G \otimes_k \bar{k}$ and $G' \otimes_k \bar{k}$ are isomorphic.
- (iii) There exists a finite separable extension K of k such that $G \otimes_k K$ and $G' \otimes_k K$ are isomorphic.

Corollary 5.9. *Let S be a nonempty scheme and $\mathcal{R}$ a root datum. The following conditions are equivalent:⁽¹⁵⁾*

- (i) There exists a pinned S -group of type $\mathcal{R}$.
- (ii) There exists an S -group of type $\mathcal{R}$.
- (iii) There exists locally for fpqc a reductive S -group of type $\mathcal{R}$.

PROOF. The task is obviously to prove (iii) $\Rightarrow$ (i). To simplify the proof, suppose there exist a faithfully flat quasi-compact morphism $S' \rightarrow S$ and a reductive S' -group G' of type $\mathcal{R}$. One may suppose G' splittable; fix a pinning E' of G' ; write $\mathcal{R} = \mathcal{R}(G', E')$. The two inverse images of (G', E') on $S'' = S' \times_S S'$ are pinned groups (G''_1, E''_1) , (G''_2, E''_2) ; one has canonical isomorphisms $p_i : \mathcal{R}(G''_i, E''_i) \xrightarrow{\sim} \mathcal{R}$, whence an isomorphism

$$p = p_2^{-1} \circ p_1 : \mathcal{R}(G''_1, E''_1) \rightarrow \mathcal{R}(G''_2, E''_2).$$

By the uniqueness theorem, there exists a unique isomorphism

$$f : (G''_1, E''_1) \xrightarrow{\sim} (G''_2, E''_2)$$

such that $\mathcal{R}(f) = p$. One therefore has a gluing datum on G' ; this is a descent datum.

Indeed, one must check a compatibility condition between the inverse images of f on S'' , but it suffices to make this check on the transforms of these arrows under the functor $\mathcal{R}$, since the latter is fully faithful. Now p is indeed a descent datum, by construction, which shows that f is also one. Since G' is affine, this descent datum is effective; since the pinning of G' is stable under the descent datum, one easily checks that there exists a pinned S -group (G, E) which gives (G', E') by base extension and which is therefore of type $\mathcal{R}$.

N.B. Naturally, in the language of fibered categories, the preceding proof becomes simpler (and comprehensible). $\square$

⁽¹⁴⁾**N.D.E.** Another proof of this corollary, not using reduction to the case of rank 2 groups, was given by M. Takeuchi ([Ta83], Th. 4.6); see also [Ja87], II 1.14.

⁽¹⁵⁾**N.D.E.** This corollary is made unnecessary by Exp. XXV, which shows the existence of a split group of type $\mathcal{R}$ over $\text{Spec}(\mathbf{Z})$, hence over any base S . (In fact, after XXV 1.3 one finds a reference to the present corollary to ensure that the reductive $\mathbf{Z}$ -group obtained is split, but this already follows from XXII 2.2; cf. N.D.E. (3) of XXV.)

Corollary 5.10. *Let S be a nonempty scheme. Let $\mathcal{R}$ be a pinned root datum such that there exists a reductive S -group of type $\mathcal{R}$. Then there exists a pinned S -group of type $\mathcal{R}$, unique up to a unique isomorphism.*

Definition 5.11. Under the preceding conditions, one will denote by $\acute{\text{E}}\text{p}_S(\mathcal{R})$ the unique pinned S -group of type $\mathcal{R}$, by $T_S(\mathcal{R})$ its canonical maximal torus, by $B_S(\mathcal{R})$ its canonical Borel subgroup, ...

If one has a morphism $S' \rightarrow S$ (S' nonempty), one may identify $\acute{\text{E}}\text{p}_{S'}(\mathcal{R})$ with $\acute{\text{E}}\text{p}_S(\mathcal{R}) \times_S S'$, etc. In particular, if $\acute{\text{E}}\text{p}_{\text{Spec}(\mathbf{Z})}(\mathcal{R})$ exists (we shall see that this is always the case), one denotes it by $\acute{\text{E}}\text{p}(\mathcal{R})$ and one has

$$\acute{\text{E}}\text{p}_S(\mathcal{R}) = \acute{\text{E}}\text{p}(\mathcal{R}) \times_{\text{Spec}(\mathbf{Z})} S.$$

One says that $\acute{\text{E}}\text{p}(\mathcal{R})$ is the *Chevalley group scheme of type $\mathcal{R}$* .

5.12. — It therefore amounts to the same thing to say that the S -group sheaf G is a reductive S -group of type $\mathcal{R}$ or to say that it is locally isomorphic (for the étale or fpqc topology) to $\acute{\text{E}}\text{p}_S(\mathcal{R})$. Likewise, by the conjugacy theorems, it amounts to the same thing to say that (G, T) is a reductive S -group of type $\mathcal{R}$ equipped with a maximal torus or that it is locally isomorphic to $(\acute{\text{E}}\text{p}_S(\mathcal{R}), T_S(\mathcal{R}))$; likewise with Borel subgroups or Killing pairs.

6. Chevalley systems

The explicit calculations of no. 3 have important numerical consequences. Let us first give the following definition:

Definition 6.1. Let S be a scheme, and (G, T, M, R) a split S -group. A *Chevalley system* of G is a family $(X_\alpha)_{\alpha \in R}$ of elements

$$X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times$$

satisfying the following condition:

(SC) for every pair $\alpha, \beta \in R$, one has

$$\text{Ad}(w_\alpha(X_\alpha))X_\beta = \pm X_{s_\alpha(\beta)}.$$

Recall (Exp. XX, 3.1) that

$$w_\alpha(X_\alpha) = \exp(X_\alpha) \exp(-X_\alpha^{-1}) \exp(X_\alpha).$$

Observe that (SC) implies in particular $X_{-\alpha} = \pm X_\alpha$ for $\alpha \in R$, by the relation (Exp. XX, 3.7) $\text{Ad}(w_\alpha(X_\alpha))X_\alpha = -X_\alpha^{-1}$.

Proposition 6.2. *Every split group has a Chevalley system. More precisely, let $(\Delta, (X'_\alpha)_{\alpha \in \Delta})$ be a pinning (1.1) of the split group (G, T, M, R) ; there then exists a Chevalley system $(X_\alpha)_{\alpha \in R}$ of G such that $X_\alpha = X'_\alpha$ for $\alpha \in \Delta$.*

Let us first show that it suffices to check condition (SC) for $\alpha \in \Delta$; for every $\alpha \in R$, there exists a sequence $\{\alpha_i\} \subset \Delta$ such that $\alpha = s_{\alpha_1} \cdots s_{\alpha_n}(\alpha_{n+1})$, whence, applying condition (SC) for each α_i ,

$$X_\alpha = \pm \text{Ad}(w_{\alpha_1}(X_{\alpha_1}) \cdots w_{\alpha_n}(X_{\alpha_n}))X_{\alpha_{n+1}}.$$

By 3.1.1 (iii), one has

$$\begin{aligned} & w_{\alpha_1}(X_{\alpha_1}) \cdots w_{\alpha_n}(X_{\alpha_n}) w_{\alpha_{n+1}}(X_{\alpha_{n+1}}) \\ & \cdot w_{\alpha_n}(X_{\alpha_n})^{-1} \cdots w_{\alpha_1}(X_{\alpha_1})^{-1} = \alpha^*(\pm 1) w_\alpha(X_\alpha). \end{aligned}$$

Now it suffices to observe that $w_{\alpha_i}(X_{\alpha_i})^{-1} = \alpha_i^*(-1) w_{\alpha_i}(X_{\alpha_i})$ and that for every pair of roots (β, γ) , one has $\beta(\gamma^*(-1)) = (-1)^{(\gamma^*, \beta)} = \pm 1$, which implies that if (SC) holds for the pairs (α_i, γ) ($\gamma \in R$), it holds for every pair (α, β) , ($\beta \in R$).

Let us now construct a system $(X_\alpha)_{\alpha \in R}$ in the following manner. For every $\alpha \in R$, choose a sequence $\{\alpha_i\} \subset \Delta$ as above and define X_α by

$$X_\alpha = \text{Ad}(w_{\alpha_1}(X_{\alpha_1}) \cdots w_{\alpha_n}(X_{\alpha_n}))X'_{\alpha_{n+1}}.$$

To check (SC), it suffices to prove:

Lemma 6.3. *Let $(G, T, M, R, \Delta, (X_\alpha)_{\alpha \in \Delta})$ be a pinned S -group; let α_i ($0 \leq i \leq n+1$) be a sequence of simple roots such that*

$$\text{int}(s_{\alpha_1} \cdots s_{\alpha_n})(\alpha_{n+1}) = \alpha_0.$$

Then

$$\text{Ad}(w_{\alpha_1} \cdots w_{\alpha_n})X_{\alpha_{n+1}} = \pm X_{\alpha_0}.$$

PROOF. Arguing as in 2.3.4 with Tits's lemma (Exp. XXI 5.6), one is reduced to checking Lemma 6.3 in the following two cases:

a) G has semisimple rank at most 2 or b) $w_{\alpha_1} \cdots w_{\alpha_n}$ is a section of T . In case a), observe that 6.3 is a consequence of 6.2 and that 6.2 was checked in part (i) of 3.1.2 (resp. 3.2.1, resp. 3.3.1, resp. 3.4.1).

It therefore remains to prove 6.3 in case b), or, equivalently, that if $\{\alpha_i\}$ is a sequence of simple roots such that $s_{\alpha_1} \cdots s_{\alpha_n} = \text{id}$, then $t = w_{\alpha_1} \cdots w_{\alpha_n}$ satisfies $\alpha(t) = \pm 1$ for every root $\alpha \in R$. By the structure of the Weyl group (Exp. XXI, 5.1), the word $s_{\alpha_1} \cdots s_{\alpha_n}$ in the free group generated by the s_α , $\alpha \in \Delta$, lies in the invariant subgroup generated by the $(s_\alpha s_\beta)^{n_{\alpha\beta}}$, $(\alpha, \beta) \in \Delta \times \Delta$. One is therefore reduced to the case where $s_{\alpha_1} \cdots s_{\alpha_n}$ has the form

$$s_{\alpha_1} \cdots s_{\alpha_i} (s_{\alpha_{i+1}} s_{\alpha_{i+2}})^{n_{\alpha_{i+1}\alpha_{i+2}}} s_{\alpha_i} \cdots s_{\alpha_1}.$$

Then one has

$$t = s_{\alpha_1} \cdots s_{\alpha_i} (t_{\alpha_{i+1}\alpha_{i+2}}),$$

and one is reduced to checking that for every pair of simple roots (α_1, α_2) and every root $\beta \in R$, one has $\beta(t_{\alpha_1\alpha_2}) = \pm 1$, which is trivial, in view of the values of $t_{\alpha_1\alpha_2}$ calculated in no. 3 (part (i) of 3.1.2, 3.2.1, 3.3.1, 3.4.1). $\square$

Proposition 6.4. *Let (G, T, M, R) be a split S -group, and $(X_\alpha)_{\alpha \in R}$ a Chevalley system of G . Let α and β be two nonproportional roots; suppose*

$$\text{long}(\alpha) \leq \text{long}(\beta), \quad \text{i.e.} \quad |(\beta^*, \alpha)| \leq |(\alpha^*, \beta)|.$$

Let q and $p-1$ be the integers ≥ 0 such that the set of roots of the form $\beta + k\alpha$, $k \in \mathbf{Z}$, is

$$\{\beta - (p-1)\alpha, \dots, \beta, \dots, \beta + q\alpha\}.$$

(Cf. Exp. XXI, 2.3.5; by loc. cit., one therefore has $-(\alpha^, \beta) = q - p + 1$.) Then the commutation relation between U_α and U_β is given by the following table (which exhausts the possible cases, since the length of the preceding root chain is $p+q-1 \leq 3$), where for each $\gamma \in R$ one writes $p_\gamma(x) = \exp(xX_\gamma)$:*

(p, q)	$p_\beta(y)p_\alpha(x)p_\beta(-y)p_\alpha(-x)$
$(-, 0)$	$= e$
$(1, 1)$	$= p_{\alpha+\beta}(\pm xy)$
$(1, 2)$	$= p_{\alpha+\beta}(\pm xy)p_{2\alpha+\beta}(\pm x^2y)$
$(1, 3)$	$= p_{\alpha+\beta}(\pm xy)p_{2\alpha+\beta}(\pm x^2y)$ $\quad p_{3\alpha+\beta}(\pm x^3y)p_{3\alpha+2\beta}(\pm x^3y^2)$
$(2, 1)$	$= p_{\alpha+\beta}(\pm 2xy)$
$(2, 2)$	$= p_{\alpha+\beta}(\pm 2xy)p_{2\alpha+\beta}(\pm 3x^2y)p_{\alpha+2\beta}(\pm 3xy^2)$
$(3, 1)$	$= p_{\alpha+\beta}(\pm 3xy).$

PROOF. By Exp. XXI, 3.5.4, there exists a system of simple roots Δ of R satisfying: $\alpha \in \Delta$ and there exist $\alpha' \in \Delta$ and $a, b \in \mathbf{Q}_+$ such that $\beta = a\alpha + b\alpha'$. Consider the pinning $(\Delta, (X_\alpha)_{\alpha \in \Delta})$ of G . The commutation relation to be checked is a relation between elements of $U_{\alpha\alpha'}$; one is therefore reduced to the explicit calculations of no. 3, and one concludes immediately by condition (SC). $\square$

Corollary 6.5. (Chevalley's rule). *Let S be a scheme, and $(X_\alpha)_{\alpha \in R}$ a Chevalley system of the split S -group G . If $\alpha, \beta, \alpha + \beta \in R$, then*

$$[X_\alpha, X_\beta] = \pm p X_{\alpha+\beta},$$

where p is the smallest integer > 0 such that $\beta - p\alpha$ is not a root.

Indeed, since the assertion is symmetric in α and β , one may suppose $\text{long}(\alpha) \leq \text{long}(\beta)$, and one is reduced to 6.4.

Corollary 6.6. *Let S be a scheme such that $6 \cdot 1_S \neq 0$ and (G, T, M, R) a split S -group. If R' is a subset of R such that*

$$(*) \quad \mathfrak{g}_{R'} = \mathfrak{t} \oplus \coprod_{\alpha \in R'} \mathfrak{g}^\alpha$$

is a Lie subalgebra of $\mathfrak{g}$, then R' is a closed subset of R (Exp. XXI, 3.1.4).

⁽¹⁶⁾Indeed, let $(X_\gamma)_{\gamma \in R_+}$ be a Chevalley system of $\mathfrak{g}$ and let $\alpha, \beta \in R'$ be such that $\alpha + \beta \in R$. By 6.5 and 6.4, one has

$$[X_\alpha, X_\beta] = \pm p X_{\alpha+\beta}, \quad \text{with } p \in \{1, 2, 3\}$$

and since neither 2 nor 3 is zero on S , this implies, by $(*)$, that $\mathfrak{g}_{R'}$ contains $\mathfrak{g}^{\alpha+\beta}$, and hence $\alpha + \beta \in R'$.⁽¹⁷⁾

6.7. — It is possible to specify the exact value of the various $\pm$ of this number, through the study of the group $\underline{\text{Norm}}_G(T)$ and, more precisely, of the “extended Weyl group”:

$$W^* = N(\mathbf{Z}), \quad \text{where } N = \underline{\text{Norm}}_{\dot{\text{E}}_{\mathbf{P}_{\mathbf{Z}}}(\mathcal{R})}(T_{\mathbf{Z}}(\mathcal{R}))$$

(cf. 5.11), which is an extension of $W(\mathcal{R})$ by an abelian group of type $(2, 2, \dots, 2)$, which is “responsible for the signs”⁽¹⁸⁾.

Remark 6.7.1. ⁽¹⁹⁾Observe that, by point (i) of 3.1.2 and 3.n.1 ($n = 2, 3, 4$), the elements w_α and w_β of $N(\mathbf{Z})$ satisfy the “braid relations”:

$$w_\alpha w_\beta \cdots = w_\beta w_\alpha \cdots \quad (n_{\alpha\beta} \text{ factors in each term}).$$

(See also [Ti66], [BLie], §IX.4, Ex. 12, and [Sp98], 9.3.2.)

⁽¹⁶⁾**N.D.E.** We have added the following proof. Observe that it suffices that 2 and 3 be nonzero on S ; for example the result holds for $S = \text{Spec}(\mathbf{Z}/6\mathbf{Z})$.

⁽¹⁷⁾**N.D.E.** On the other hand, let us point out that if $2 = 0$ on S and R is of type C_n , then the set R' of short roots (which is a root system of type D_n) is not closed in R , but is a symmetric subset of type (R) of R (cf. XXII 5.4.2 and 5.4.10), i.e., there corresponds to it a subgroup $H_{R'}$ of type (R) of G with reductive fibers: this is in particular the case for the natural embedding (in characteristic 2) of $\text{SO}(2n)$ in $\text{Sp}(2n)$. Likewise, if $2 = 0$ on S and R is of type F_4 (resp. if $3 = 0$ on S and R is of type G_2), the set R' of short roots (which is a root system of type D_4 (resp. A_2)) is not closed in R , but corresponds to a subgroup $H_{R'}$ of type (R) of G with reductive fibers.

* Cf. J. Tits: *Sur les constantes de structure et le théorème d'existence des algèbres de Lie semi-simples*, Publ. Math. I.H.É.S. **31** (1966), 21–58.

⁽¹⁸⁾**N.D.E.** See also [Ti66].

⁽¹⁹⁾**N.D.E.** We have added this remark.

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